Repro Samples Method for a Performance Guaranteed Inference in General and Irregular Inference Problems

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Abstract

Rapid advances in data science require fundamentally new approaches to address prevalent inference problems for which regularity conditions and the large-sample central limit theorem do not apply, such as those involving discrete or non-numerical parameters or non-numerical data. This article presents an innovative and effective framework, called *repro samples method*, to conduct statistical inference in general settings, including the complex problems described above. The development leverages Fisher inversion techniques and artificial samples generated to mimic the observed data, and is broadly applicable and supported by rigorous theories. For commonly encountered irregular inference problems involving mixed types of (discrete/non-numerical and continuous) parameters, a three-step procedure is proposed to dissect the complicated inference task into manageable steps. The article also develops a unique matching scheme that turns parameter discreteness from an obstacle for forming inferential theories into an advantage for improving computational efficiency. The effectiveness of the proposed methodology is demonstrated using examples, including a case study on inference for a Gaussian mixture model with an unknown number of components that resolves a long-standing inference problem. Real data and simulation studies, with comparisons to existing approaches, demonstrate the superior performance of the proposed method.

Key words: Artificial data; Discrete or non-numerical parameters; Gaussian mixture; Generative model; Inversion methods; Parametric and nonparametric models

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1 Introduction

The approaches of creating artificial data to help assess uncertainty for inference have proven to be an effective method in the literature; see, e.g., Efron and Tibshirani (1993); Robert (2016); Hannig et al. (2016); Dalmasso et al. (2024), and references therein. Except for a few recent works, most of the approaches, however, still rely on large-sample central limit theorem (CLT) to justify their inference validity. Their applicability to more complex problems, especially those involving discrete or non-numerical parameters and non-numerical data is limited. In this article, we develop a fundamentally new and wide-reaching artificial-sample-inspired inferential framework, with supporting theories, for these complex problems and more. The framework, called *repro samples method*, does not need to rely on CLT or likelihood functions. It is especially effective for difficult *irregular inference problems*. Here, following Wasserman et al. (2020), the irregular problems refer to those "highly non-trivial" and "complex" setups to which the regularity assumptions for large-sample theories do not apply. Although our focus is mostly on finite-sample inference, the proposed framework has direct extensions to the situations where the large-sample theorem holds as well.

Suppose sample data $\boldsymbol{Y} \in \mathcal{Y}$ are generated from a model (distribution) $\boldsymbol{Y}|\boldsymbol{\theta} \sim F_\theta(\cdot)$, where $\boldsymbol{\theta} \in \Theta$ is model parameters, which can either be numerical, non-numerical, or a mixture of both, or even functions or model indices. In most publications $\boldsymbol{Y}$ typically have n data points $\boldsymbol{Y} = (Y_1, \ldots, Y_n)^\top$, but we allow $n = 1$ or also $\boldsymbol{Y}$ to be a summary or function of $\boldsymbol{Y}$; they can also be non-numerical types (e.g., image; voice; object; etc.). We assume that we know how to simulate $\boldsymbol{Y}$ from $F_\theta(\cdot)$, given $\boldsymbol{\theta}$. Often, $\boldsymbol{Y}|\boldsymbol{\theta} \sim F_\theta(\cdot)$ can be re-expressed as: in the form of *a generative model*, which we adopt in this paper:

$$\boldsymbol{Y} = G(\boldsymbol{\theta}, \boldsymbol{U}), \tag{1}$$

where $G(\cdot, \cdot)$ is a known mapping from $\Theta \times \mathcal{U} \mapsto \mathcal{Y}$ and $\boldsymbol{U} = (U_1, \ldots U_r)^\top \in \mathcal{U}$, for some $r > 0$, is a random vector whose distribution, say $D_U(\cdot)$, is free of $\boldsymbol{\theta}$.

The generative model (1) is frequently used in modern data science literature (e.g., Dalmasso et al., 2024, among others) and very general. It is natural for problems where $\boldsymbol{Y}$ are generated (simulated) through an algorithm. The classical statistical model specification using a family of (parametric) density functions $Y_i \sim f_\theta(y)$, for $i = 1, \ldots, n$, can also be re-expressed in this form $Y_i = G(\theta, U_i)$, with $G(\theta, \cdot) = F_\theta^{-1}(\cdot)$, $F_\theta(\cdot)$ being the cumulative distribution function and $U_i \sim U(0,1)$. Additionally, for model distributions with a probability measure on Borel spaces, it is always possible to separate the model parameters from the stochastic component — By the Borel isomorphism theorem (Kallenberg, 2021, Theorem 4.19), any probability measure on a Borel space can be represented by a random element on $[0,1]$, with the corresponding distribution generated from a $U(0,1)$ random variable. Thus, for any model with a corresponding probability measure on a Borel space, its sample random variable(s) $\boldsymbol{Y}$ can be expressed as $\boldsymbol{Y} = \Psi(U)$, where $U \sim U(0,1)$, $\Psi(\cdot)$ is a deterministic measurable bijection from $[0,1]$ to $\mathcal{Y}$, and the randomness of $\boldsymbol{Y}$ is entirely represented by the random component U. The model parameters are typically characteristics (features) of the mapping function Ψ, $\boldsymbol{\theta} = \boldsymbol{\theta}(\Psi)$, which can be any types and may even be Ψ itself. In this paper, for notational convenience and to facilitate the inversion techniques to be used, we explicitly separate the model parameters and the random components by rewriting $\boldsymbol{Y} = \Psi(U)$ as $\boldsymbol{Y} = G(\theta, \boldsymbol{U})$, where, to better accommodate practical use, we also allow $\boldsymbol{U} \in \mathcal{U} \subset \mathbb{R}^r$ to be a $r \times 1$ random vector from a known distribution on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For ease of discussion, we further assume that the target model parameter $\boldsymbol{\theta} = \boldsymbol{\theta}(\Psi)$ is well-defined and identifiable, and that the sample data $\boldsymbol{Y}$ provide enough information to make inference about $\boldsymbol{\theta}$. An additional discussion on model identifiability is in Section 6.

Let $\boldsymbol{y}_{obs} = G(\boldsymbol{\theta}_0, \boldsymbol{u}^{rel})$ be the realized observation with true parameter value $\boldsymbol{\theta}_0$ and $\boldsymbol{u}^{rel}$ be the corresponding (unobserved) realization of $\boldsymbol{U}$. The repro samples framework uses a simple yet fundamental idea: study artificial samples obtained by mimicking the sampling mechanism; then use them to help quantify inference uncertainty. Particularly, for any $\boldsymbol{\theta} \in \Theta$ and a copy of artificial $\boldsymbol{u}^* \sim \boldsymbol{U}$, we can use (1) to get an artificial sample $\boldsymbol{y}^* = G(\boldsymbol{\theta}, \boldsymbol{u}^*)$, which we refer to as a *repro sample*. If $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ and $\boldsymbol{u}^*$ matches $\boldsymbol{u}^{rel}$, the corresponding $\boldsymbol{y}^*$ matches $\boldsymbol{y}_{obs}$. Inversely, if $\boldsymbol{y}_{obs} = \boldsymbol{y}^*$, an artificial sample generated using $\boldsymbol{\theta}$ and a likely realization $\boldsymbol{u}^*$ from $\boldsymbol{U}$, then we cannot dismiss the possibility that such a $\boldsymbol{\theta}$ equals $\boldsymbol{\theta}_0$.

The above (inversion) idea can be traced back to R.A. Fisher in which one can solve a model equation to obtain $\boldsymbol{\theta}_0$ if both $\boldsymbol{y}_{obs}$ and $\boldsymbol{u}^{rel}$ were completely given (known). However, Fisher's continuously treating the unknown realization $\boldsymbol{u}^{rel}$ as random leads to unsolvable obstacles and his version of fiducial inference being viewed as his "biggest blunder" (e.g., Efron 1998; Hannig et al. 2016; Thornton and Xie 2024). In this paper, we will use two inversion ideas and artificial samples $\boldsymbol{u}^*$ or $\boldsymbol{y}^*$ to develop a general *frequentist* repro samples framework to construct level-$(1-\alpha)$ confidence sets, $\alpha \in (0,1)$, for parameters of interest. The development covers both regular and irregular inference problems under both finite and large-sample settings. We show that the constructed confidence sets have a theoretically guaranteed frequency coverage rate. We also show that, for any confidence set constructed by inverting a Neyman-Pearson test, the proposed method can always construct either the same or a better confidence set. We further provide discussions on handling nuisance parameters.

We specifically focus on complex and difficult inference problems that involve mixed types of parameters $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}^\top)^\top$, where η are discrete or non-numerical and, when given η, the remaining parameters $\boldsymbol{\beta}$ may be handled by a regular approach. An effective three-step approach is proposed to dissect the difficult inference problems into manageable steps. Moreover, the discrete nature of η, often an obstacle for regular inference theory, becomes a beneficial feature to help greatly improve computational and inferential efficiency within our framework. This development is applied to a case study example of a Gaussian mixture model with an unknown number of components, where we resolve a long-standing open question on quantifying uncertainty in estimating the discrete number of components and associated parameters. Note that, for inference problems involving discrete or non-numerical parameters, their point estimators, if exist, do not provide any uncertainty quantification. The often-used bootstrap approaches lack theoretical support. Bayesian methods may construct credible sets, but the sets are highly sensitive to the prior choices and their frequency coverage performance is typically poor even when the sample sizes are large (Hastie and Green, 2012). The difficulties in these bootstrap and Bayesian methods are from the fact that the CLT and Bernstein-von-Mises theorem do not apply for estimators of the discrete or non-numerical parameters. On the contrary, our repro samples method does not have such constraints.

Major contributions and significance:

1. We develop a new and broadly applicable inferential framework to construct performance-guaranteed confidence sets for parameters of interest under generative model settings. In addition to conventional inference problems, it is particularly useful for irregular inference problems with discrete or non-numerical parameters and non-numerical data.
2. We show that a confidence set obtained by the repro samples method is either the same as or smaller than that obtained by inverting the Neyman-Pearson test. We obtain an optimality result that an optimal Neyman confidence set corresponds to an optimal confidence set by the repro samples method; we also provide examples that the repro samples method improves the result by the Neyman method when it is not optimal.
3. The repro samples method provides explicit solutions in many cases. In other cases when explicit expressions are unavailable, we provide a practical guideline and an generic algorithm, often with Monte-Carlo simulation, to compute confidence sets. Two techniques are provided to significantly reduce the computing costs: (a) for discrete parameters, we utilize a unique many-to-one inversion mapping to develop a novel data-driven approach, with supporting theories, to reduce the parameters' search space; (b) for continuous parameters, we adopt a quantile regression technique used by Dalmasso et al. (2024) to avoid grid search.
4. We provide a general profiling method to handle nuisance parameters for finite-sample inference in the repro samples framework. Unlike the conventional 'hybrid' or 'likelihood profiling' methods that often rely on asymptotics (e.g., Chuang and Lai, 2000; Dalmasso

et al., 2024), the proposed method is developed based on specifically constructed p-values, and it guarantees the finite-sample performance of the repro samples confidence sets.

5. In a case study illustration, we provide a solution to an open and "highly nontrivial" inference problem on how to quantify the uncertainty in estimating the unknown number of components (say τ_0) and the associated parameters in a Gaussian mixture model (cf., Wasserman et al., 2020). Numerical studies based on a real data setting show that the repro samples method is the only approach among existing frequentist and Bayesian approaches that can cover τ_0 with the desired accuracy and also be effective for other parameters.

Relation to other work: Besides the classical Neyman-Pearson test and the recent work on *simulation-based inference* (SBI) (Dalmasso et al., 2024), the repro samples method stems from and is closely related to several simulation-based approaches across Bayesian, frequentist and fiducial (BFF) paradigms (Berger et al., 2020), namely, *Bootstrap* (Efron and Tibshirani, 1993), *Approximate Bayesian Computation* (ABC) (Robert, 2016), *Generalized fiducial Inference* (GFI) (Hannig et al., 2016) and *Inferential Model* (IM) (Martin and Liu, 2015). Specifically, the two key inversion techniques, the direct *Fisher inversion* in Section 3.2 and its generalization to operate on Borel sets, the *Fisher–Dempster inversion* in Section 2, are also a key technique used in GFI (Hannig et al., 2016) and IM (Martin and Liu, 2015), respectively, although our formulations and applications are a little different, and our repro samples method is fully frequentist and does not involve any fiducial justification or use of Dempster–Shafer calculus. Moreover, bootstrap, GFI and most other large-sample methods rely on CLT, except in special cases; they often cannot handle irregular inference problems involving discrete/non-numerical parameters or structures. Furthermore, the required sufficiency in ABC, the dependence of BvM-type theorems (thus CLT) for inference justification in GFI and ABC (Hannig et al., 2016; Li and Fearnhead, 2018) and the unaddressed ϵ (approximation error) question (Li and Fearnhead, 2018) in ABC and GFI have hampered the use of these approaches in practice. Finally, *universal inference* (Wasserman et al., 2020) is another novel framework for performance-guaranteed inferences without classical regularity conditions, although it relies on a Markov inequality to justify its validity (test size) at some expanse of power (Dunn et al., 2022; Tse and Davison, 2022). A Gaussian mixture example is also studied in Wasserman et al. (2020) but, unlike our method, it cannot provide a two-sided confidence set for τ_0 and has less power than ours. Due to space limit, we provide detailed reviews and comparisons of these related methods in Appendix E.

Organization of the remaining sections: Section 2 develops a basic yet general framework of the repro samples method with supporting theories. A connection and comparison to the classical Neyman inversion approach is also provided. Section 3 contains further developments on handling nuisance parameters, constructing a data-driven candidate set to improve computing efficiency, and developing an effective three-step procedure tailored for problems with mixed types of parameters. A practical guideline for the use of the repro samples method is also provided. Section 4 is a case study that addresses a long-standing open inference problem in the Gaussian mixture model. Section 5 contains real data and simulation studies. Section 6 includes further discussions. Due to space limit, proofs of all theorems, as well as additional algorithms, examples, and numerical results, are provided in the appendices.

2 A general inference framework by repro samples

Let $\boldsymbol{U} \in \mathcal{U} \subset \mathbb{R}^r$ be a measurable random vector on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\{\boldsymbol{U} \leq \boldsymbol{u}\} = \{\omega \in \Omega : \boldsymbol{U}(\omega) \leq \boldsymbol{u}\}$. The randomness of $\boldsymbol{Y} = \boldsymbol{Y}(\omega)$ is from $\boldsymbol{U} = \boldsymbol{U}(\omega)$. For a given $\boldsymbol{\theta}$, we call a simulated $\boldsymbol{u}^* \sim \boldsymbol{U}$ a *repro sample* of $\boldsymbol{u}^{rel}$ and $\boldsymbol{y}^* = G(\boldsymbol{\theta}, \boldsymbol{u}^*)$ a *repro sample* of $\boldsymbol{y}_{obs}$.

2.1 Basic version: a level-(1 - α) confidence set by Fisher-Dempster inversion

Let $T(\cdot,\cdot)$ be a (user specified) mapping function from $\mathcal{U} \times \Theta \to \mathcal{T} \subseteq \mathbb{R}^q$, for some $q \leq n$, which we refer to as a *nuclear mapping*. We first quantify the uncertainty of the random $\boldsymbol{U}$ via the function $T(\cdot,\cdot)$ and a Borel set $B_{1-\alpha}(\boldsymbol{\theta}) \subset \mathcal{T}$:

$$\mathbb{P}\left\{T(\boldsymbol{U}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\right\} \geq 1 - \alpha, \tag{2}$$

for a given $\boldsymbol{\theta}$. Then, let "s.t." be an abbreviation for "such that," we construct a set in Θ:

$$\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) = \big\{ \boldsymbol{\theta} : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } \boldsymbol{y}_{obs} = G(\boldsymbol{\theta}, \boldsymbol{u}^*),\, T(\boldsymbol{u}^*, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta}) \big\} \subset \Theta. \tag{3}$$

That is, for a potential $\boldsymbol{\theta} \in \Theta$, if there exists a realization $\boldsymbol{u}^*$, with $T(\boldsymbol{u}^*, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})$, such that the repro sample $\boldsymbol{y}^* = G(\boldsymbol{\theta}, \boldsymbol{u}^*)$ matches with $\boldsymbol{y}_{obs}$, then we keep this $\boldsymbol{\theta}$ in set $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$. Here, the observed $\boldsymbol{y}_{obs}$ can be produced by any $\boldsymbol{\theta} \in \Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$, with a corresponding potential realization $\boldsymbol{u}^*$ (satisfying (2)), so we cannot rule out the possibility of these $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. Theorem 1 below states that $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ is a level $1-\alpha$ confidence set in both finite-sample and large-sample (or other) approximations, followed by a corollary on hypothesis testing.

Theorem 1. *Assume model (1) holds with $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. If inequality (2) holds exactly for any fixed $\boldsymbol{\theta}$, then the following inequality holds exactly*

$$\mathbb{P}\left\{\boldsymbol{\theta}_0 \in \Gamma_{1-\alpha}(\boldsymbol{Y})\right\} \geq 1-\alpha \;\; \textit{for } 0 < \alpha < 1. \tag{4}$$

Furthermore, if the inequality (2) holds approximately with $\mathbb{P}\big\{T(\boldsymbol{U}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\big\} \geq \alpha\{1 + o(\delta')\}$, then (4) holds approximately with $\mathbb{P}\left\{\boldsymbol{\theta}_0 \in \Gamma_{1-\alpha}(\boldsymbol{Y})\right\} \geq \alpha\{1 + o(\delta')\}$, for $0 < \alpha < 1$, where $\delta' > 0$ is a small value that may or may not depend on sample size n.

Corollary 1. *For $\Theta_0 \subsetneq \Theta$ and a test $H_0 : \boldsymbol{\theta} \in \Theta_0$ vs $H_1 : \boldsymbol{\theta} \notin \Theta_0$, we can define a p-value*

$$p(\boldsymbol{y}_{obs}) = 1 - \inf\nolimits_{\boldsymbol{\theta} \in \Theta_0} \left[\inf\left\{1 - \gamma : \boldsymbol{\theta} \in \Gamma_{1-\gamma}(\boldsymbol{y}_{obs})\right\}\right],$$

where $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ is by (3). Rejecting H_0 when $p(\boldsymbol{y}_{obs}) \leq \alpha$ leads to a size α test, $0 < \alpha < 1$.

In Theorem 1, δ' may depend on n with $\delta' \to 0$ as $n \to 0$ for a large-sample approximation, but there are also examples in which δ' does not involve n. For example, if $Y|\theta = \lambda \sim$ Poisson(λ), then $U = \frac{Y-\lambda}{\sqrt{\lambda}} \to N(0,1)$ for a large λ. So, taking $T(U, \lambda) = U$, we have $\mathbb{P}\{T(U, \lambda) \in B\} = \int_{t \in B} \phi(t)dt\{1 + o(\frac{1}{\lambda})\}$ for any set B; thus $\delta' = \frac{1}{\lambda}$. Here, $\phi(t)$ is $N(0,1)$ density function. Also, in (2) and (4), we keep the inequalities "$\geq$" to cover the situations when either $\boldsymbol{Y}$ or $\boldsymbol{\theta}_0$ is discrete. These two "$\geq$" can be replaced by "$=$" in other situations and the method does not suffer systemic power loss (unlike the universal inference method — see an elaboration in Appendix E.3.1). Further discussions on optimality in comparison with Neyman-Pearson test is in Section 2.2, where we show the repro samples method is broader and can achieve the same or better results. Asymptotic discussions may help but are not needed for efficiency.

In the repro samples development, the role of the nuclear mapping $T(\cdot,\cdot)$ is similar to, but broader than, that of a test statistic in the classical Neyman-Pearson framework. We assume throughout the paper that $T(\cdot,\cdot)$ is provided (user-specified), and a general guide on its choice is given in Section 3.4. Regardless of its choice, the repro samples method always has the desired coverage rate, but the power may be impacted. Two obvious special cases are: (i) $T(\boldsymbol{U},\boldsymbol{\theta}) = \boldsymbol{U}$, unrelated to $\boldsymbol{\theta}$, and (ii) $T(\boldsymbol{U},\boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{Y},\boldsymbol{\theta})$ being a test statistic. The special case (i) is closely related to the IM development of Martin and Liu (2013), although IM attempts to achieve a higher level goal of producing a probabilistic inference for $\boldsymbol{\theta}$ that requires random sets (not a single fixed Borel set) and use of imprecise probability system *Dempster-Shafer calculus*. Our development also allows $T(\boldsymbol{U},\boldsymbol{\theta})$ depending on $\boldsymbol{\theta}$ under consideration, and it provides a great flexibility which is necessary for many complex inference problems. The special case (ii) is closely related to the classical Neyman inversion method, although $T(\cdot,\cdot)$ does not need to be a test statistic and is more general. Section 2.2 next investigates in detail the special case (ii) and its connection to the classical Neyman-Pearson method.

To illustrate the proposed repro samples method, we consider below a binomial example.

Example 1 (Binomial sample with success probability θ_0)**.** *Assume $Y \sim Binomial(r,\theta)$ $0<\theta<1$. In the form of (1), $Y = \sum_{i=1}^r \mathbf{1}(U_i \le \theta_0)$ for $U_1,\ldots,U_r \sim U(0,1)$. Correspondingly, $y_{obs} = \sum_{i=1}^r \mathbf{1}(u_i^{rel} \le \theta_0)$, for a realized y_{obs} from θ_0 and $\boldsymbol{u}^{rel} = (u_1^{rel},\ldots,u_r^{rel})$. We consider a nuclear mapping function $T(\boldsymbol{u},\theta) = \sum_{i=1}^r \mathbf{1}(u_i \le \theta)$, so $T(\boldsymbol{U},\theta) = \sum_{i=1}^r \mathbf{1}(U_i \le \theta) \sim Binomial(r,\theta)$, for each given $\theta \in \Theta = (0,1)$. Let $B_{1-\alpha}(\theta) = [a_L(\theta), a_U(\theta)]$ in (2) be the shortest interval whose bounds are (i,j) pairs in $\{(i,j) : \sum_{k=i}^j \binom{r}{k}\theta^k(1-\theta)^{(r-k)} \ge 1-\alpha\}$; i.e.,*

$$(a_L(\theta), a_U(\theta)) = \arg\min_{\left\{(i,j):\sum_{k=i}^j \binom{r}{k}\theta^k(1-\theta)^{(r-k)} \ge 1-\alpha\right\}} |j-i|. \tag{5}$$

Equation (2) can be directly verified and, after simplification, the confidence set (3) becomes

$$\Gamma_{1-\alpha}(y_{obs}) \quad = \quad \left\{\theta \middle| a_L(\theta) \le y_{obs} \le a_U(\theta)\right\}. \tag{6}$$

Due to space limit, the derivation to get (6) and also a numerical study are placed in Ap-

pendix A.3. The numerical study shows the finite-sample interval (6) has the desired coverage rate in all settings (including $n\theta_0 < 5$ cases). It performs similarly to (slightly better than) Stern's exact method and outperforms other four existing, both exact and asymptotic, methods.

2.2 Test statistic as nuclear mapping function and Neyman inversion of test

Under the classical hypothesis testing setup $H_0 : \boldsymbol{\theta}_0 = \boldsymbol{\theta}$ vs $H_1 : \boldsymbol{\theta}_0 \neq \boldsymbol{\theta}$, we often construct a test statistic, say $\widetilde{T}(\boldsymbol{y}_{obs}, \boldsymbol{\theta})$, and assume we know the distribution of $\widetilde{T}(\boldsymbol{Y}, \boldsymbol{\theta})$ for $\boldsymbol{Y} = G(\boldsymbol{\theta}, \boldsymbol{U})$ generated under H_0. By the classical testing method, we can derive a level $1-\alpha$ acceptance region $A_{1-\alpha}(\boldsymbol{\theta}) = \big\{\boldsymbol{y}_{obs} \big| \widetilde{T}(\boldsymbol{y}_{obs}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\big\}$, where set $B_{1-\alpha}(\boldsymbol{\theta})$ satisfies $\mathbb{P}\big\{\widetilde{T}(\boldsymbol{Y}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\big\} \geq 1-\alpha$. Then, by the *Neyman's inversion* (test duality) method, a level $1-\alpha$ confidence set is $\widetilde{\Gamma}_{1-\alpha}(\boldsymbol{y}_{obs}) = \{\boldsymbol{\theta} : \boldsymbol{y}_{obs} \in A_{1-\alpha}(\boldsymbol{\theta})\} = \Big\{\boldsymbol{\theta} : \widetilde{T}(\boldsymbol{y}_{obs}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\Big\}$.

Now suppose our nuclear mapping $T(\boldsymbol{u}, \boldsymbol{\theta})$ is defined through the same test statistic as:

$$T(\boldsymbol{U}, \boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{Y}, \boldsymbol{\theta}) = \widetilde{T}(G(\boldsymbol{\theta}, \boldsymbol{U}), \boldsymbol{\theta}), \tag{7}$$

where $\boldsymbol{Y} = G(\boldsymbol{\theta}, \boldsymbol{U})$ is from the same $\boldsymbol{\theta}$. Since $\mathbb{P}\left\{T(\boldsymbol{U}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\right\} = \mathbb{P}\big\{\widetilde{T}(\boldsymbol{Y}, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\big\} \geq 1-\alpha$ for the same $B_{1-\alpha}(\boldsymbol{\theta})$, the repro samples confidence set (3) becomes:

$$\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) = \Big\{\boldsymbol{\theta} : \exists \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } \boldsymbol{y}_{obs} = G(\boldsymbol{\theta}, \boldsymbol{u}^*), \widetilde{T}(G(\boldsymbol{\theta}, \boldsymbol{u}^*), \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})\Big\}.$$

Theorem 2 below states that $\widetilde{\Gamma}_{1-\alpha}(\boldsymbol{y}_{obs})$ by Neyman inversion contains the above $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$.

Theorem 2. *If the nuclear mapping function is defined through a test statistic as in (7), then we have $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) \subseteq \widetilde{\Gamma}_{1-\alpha}(\boldsymbol{y}_{obs})$. The two sets are equal $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) = \widetilde{\Gamma}_{1-\alpha}(\boldsymbol{y}_{obs})$, when $\widetilde{\Gamma}_{1-\alpha}(\boldsymbol{y}_{obs}) \subseteq \big\{\, \boldsymbol{\theta} : \boldsymbol{y}_{obs} = G(\boldsymbol{\theta}, \boldsymbol{u}^*), \exists\, \boldsymbol{u}^* \in \mathcal{U}\big\}$.*

It is possible that $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) \subsetneq \widetilde{\Gamma}_{1-\alpha}(\boldsymbol{y}_{obs})$ strictly. The following is such an example.

Example 2. *Let $\boldsymbol{y}_{obs} = (y_1, \ldots, y_n)^\top$ be an observed sample with θ_0 from the model $Y_i = \theta + U_i$, for $U_i \sim U(-1, 1)$, $i = 1, \ldots, n$, and $\Theta = \mathcal{Y} = (-\infty, \infty)$. An unbiased (also $\sqrt{n}$ consistent) point estimator is $\bar{Y} = \frac{1}{n}\sum_{i=1}^n Y_i$. Suppose we use the test statistic $\widetilde{T}(\boldsymbol{y}_{obs}, \theta) = \bar{y} - \theta$. Since $n\{(\bar{Y} - \theta) + 1\}/2 = \sum_{i=1}^n \frac{U_i + 1}{2}$ follows an Irwin-Hall distribution under $H_0 : \theta_0 = \theta$, a level 95% confidence interval by the Neyman inversion method is $\widetilde{\Gamma}_{.95}(\boldsymbol{y}_{obs}) = (\bar{y} - \frac{2}{n}q_{.975} + 1, \bar{y} + \frac{2}{n}q_{.975} - 1)$. Here, $q_{.975}$ is the 97.5% quantile of the Irwin-Hall distribution. If we use the* *repro samples method, our confidence interval* $\Gamma_{.95}(\boldsymbol{y}_{obs}) = \big\{\theta : y_i = \theta + u_i^*, i = 1, \ldots, n;\ \bar{y} \in (\theta - \frac{2}{n}q_{.975} + 1, \theta + \frac{2}{n}q_{.975} - 1); \exists \boldsymbol{u}^* \in (-1,1)^n\big\} = \{\theta : \theta \in \cap_{i=1}^n (y_i - 1, y_i + 1)\} \cap \widetilde{\Gamma}_{.95}(\boldsymbol{y}_{obs})$*. Since the constraint* $\{\theta \in \cap_{i=1}^n (y_i - 1, y_i + 1)\}$ *is often in tack, the strictly smaller statement* $\Gamma_{.95}(\boldsymbol{y}_{obs}) \subsetneq \widetilde{\Gamma}_{.95}(\boldsymbol{y}_{obs})$ *holds for many realizations of* $\boldsymbol{y}_{obs}$*. Our numerical study (due to space limit placed in Appendix A.4) also confirms this conclusion empirically — both intervals have desired empirical coverage rates* 95.1%*, but among the* $1,000$ *repetitions,* 219 *times* $\Gamma_{.95}(\boldsymbol{y}_{obs}) = \widetilde{\Gamma}_{.95}(\boldsymbol{y}_{obs})$ *and* 781 *times* $\Gamma_{.95}(\boldsymbol{y}_{obs}) \subsetneq \widetilde{\Gamma}_{.95}(\boldsymbol{y}_{obs})$*. The average length of the repro samples intervals* $|\Gamma_{.95}(\boldsymbol{y}_{obs})| = 0.915$ *is shorter than that of* $|\widetilde{\Gamma}_{.95}(\boldsymbol{y}_{obs})| = 1.292$*.*

There are two immediate implications from the special case with a nuclear mapping $T(\cdot,\cdot)$ defined via a test statistic as in (7). First, by Theorem 2,the repro samples confidence set $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ is never worse than the one obtained by the classical testing approach, thus is more desirable. Second, if the test statistic is optimal (in the sense it leads to a powerful test or an optimal confidence set), the corresponding repro samples confidence set $\Gamma(\boldsymbol{y}_{obs})$ is also optimal. Corollary 2 below is the formal statement, where a level $1-\alpha$ *uniformly most accurate* (UMA) confidence set (a.k.a. *Neyman shortest*) refers to a level $1-\alpha$ confidence set that minimizes the probability of false coverage (i.e., probability of covering a wrong parameter value) over a class of level $1-\alpha$ confidence sets (Casella and Berger, 1990, §9.3.2).

Corollary 2. *(a) If a test statistic* $\widetilde{T}(\boldsymbol{y}_{obs}, \boldsymbol{\theta})$ *corresponds to the uniformly most powerful test and* $\widetilde{\Gamma}(\boldsymbol{y}_{obs})$ *is a level* $1-\alpha$ *UMA confidence set, then the set* $\Gamma(\boldsymbol{y}_{obs})$ *by the corresponding repro sample method is also a level* $1-\alpha$ *UMA confidence set.*
(b) If a test statistic $\widetilde{T}(\boldsymbol{y}_{obs}, \boldsymbol{\theta})$ *corresponds to the uniformly most powerful unbiased test and* $\widetilde{\Gamma}(\boldsymbol{y}_{obs})$ *is a level* $1-\alpha$ *UMA unbiased confidence set, then the confidence set* $\Gamma(\boldsymbol{y}_{obs})$ *by the corresponding repro sample method is also a level* $1-\alpha$ *UMA unbiased confidence set.*

Finally, we would like to stress that the nuclear mapping $T(\cdot,\cdot)$ does not need to be a test statistic. One such example, on inference for a nonparametric quantile with noise-perturbed observations (for privacy consideration), is provided in Example A3 of Appendix A.4, in which a nuclear mapping $T(\cdot,\cdot)$ is readily available but an effective test statistic is difficult to construct. Furthermore, Neyman-Pearson lemma suggests that the likelihood ratio test (LRT) is uniformly most powerful for a simple-versus-simple hypothesis, but it does not necessarily hold for a two-sided test. Thus, an interval constructed by inverting an LRT is not necessarily optimal. Example A4 of Appendix A.4, a continuation of Example 2, demonstrates that the LRT confidence interval is not optimal, and a repro samples interval can improve it.

2.3 Beyond a fully specified generative model and defining oracle parameters

The repro samples method is also applicable even in cases when model (1) is not fully given. For instance, consider a nonparametric inference problem on the ζ-th quantile of an unknown distribution F, say, $\theta_0 = F^{-1}(\zeta)$. Since sample $Y \sim F$, we have $I(Y < \theta_0) \sim$ Bernoulli(ζ) and equation $I(Y < \theta_0) = U$, where $U \sim$ Bernoulli(ζ). Now suppose we observe data $\boldsymbol{y}_{obs} = (y_1^{obs}, \ldots, y_n^{obs})^\top$, for a fixed n, then the equation (treated as a given model specification) becomes

$$\sum\nolimits_{i=1}^{n} I(Y_i - \theta_0 < 0) - \sum\nolimits_{i=1}^{n} U_i = 0, \tag{8}$$

where $Y_i \overset{iid}{\sim} F$ and $U_i \overset{iid}{\sim}$ Bernoulli(ζ). Equation (8) cannot be re-expressed in the form of (1). Although based on (8), we cannot generate a repro sample $\boldsymbol{y}^*$ that is directly comparable to the observed $\boldsymbol{y}_{obs}$, we can still use the repro samples method. Particularly, let us consider a *generalized generative model*, of which (8) is a special case:

$$g(\boldsymbol{Y}, \boldsymbol{\theta}, \boldsymbol{U}) = \boldsymbol{0}, \tag{9}$$

where $(\boldsymbol{Y}, \boldsymbol{\theta}, \boldsymbol{U})$ is the same as in (1) and g is a given function from $\mathcal{Y} \times \Theta \times \mathcal{U} \to \mathbb{R}^s$. The realization version is $g(\boldsymbol{y}_{obs}, \boldsymbol{\theta}_0, \boldsymbol{u}^{rel}) = \boldsymbol{0}$. In this case, we modify $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ in (3) as follows:

$$\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) = \big\{ \boldsymbol{\theta} : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } g(\boldsymbol{y}_{obs}, \boldsymbol{\theta}, \boldsymbol{u}^*) = 0, T(\boldsymbol{u}^*, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta}) \big\} \subset \Theta. \tag{10}$$

A special case of (9) that is relevant to machine learning and modern data science research is: Irrespective of the assumed model structure, completely given or not, we only assume that the sample data $\boldsymbol{Y}$ contain sufficient information to recover our target model parameter $\boldsymbol{\theta}$, in the sense that the parameter $\boldsymbol{\theta}$ can be recovered by an algorithm when given $\boldsymbol{Y}$ and $\boldsymbol{U}$:

$$\boldsymbol{\theta} = H(\boldsymbol{Y}, \boldsymbol{U}). \tag{11}$$

That is, when given observed $\boldsymbol{y}_{obs}$ and its corresponding realized $\boldsymbol{u}^{rel}$, we can express our target parameter $\boldsymbol{\theta}_0$ as $\boldsymbol{\theta}_0 = H(\boldsymbol{y}_{obs}, \boldsymbol{u}^{rel})$. Here, $H(\cdot,\cdot)$ is a given algorithm or function. Under the generative model (1), for instance, $H(\cdot,\cdot)$ is typically an inversion algorithm that solves for

$$\theta_0 = H(\boldsymbol{y}_{obs}, \boldsymbol{u}^{rel}) \stackrel{\text{def}}{=} \arg\min_\theta L\big(\boldsymbol{y}_{obs}, G(\boldsymbol{\theta}, \boldsymbol{u}^{rel})\big), \tag{12}$$

where $L\big(\boldsymbol{y}, \boldsymbol{y}'\big)$ is a loss function that measures the difference between two copies of data $\boldsymbol{y}$ and $\boldsymbol{y}'$. Similarly, under (9), $H(\cdot,\cdot)$ is the algorithm that solves $g(\boldsymbol{y}_{obs}, \boldsymbol{\theta}, \boldsymbol{u}^{rel}) = \boldsymbol{0}$ for $\boldsymbol{\theta}_0$, though mathematically (11) is a special case of (9) with $g(\boldsymbol{y}, \boldsymbol{\theta}, \boldsymbol{u}) \stackrel{\text{def}}{=} H(\boldsymbol{y}, \boldsymbol{u}) - \boldsymbol{\theta}$. Under model (11), we can modify the repro sample confidence set $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ in (3) as:

$$\Gamma_{1-\alpha}(\boldsymbol{y}_{obs}) = \big\{\, \boldsymbol{\theta} = H(\boldsymbol{y}_{obs}, \boldsymbol{u}^*) : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } T(\boldsymbol{u}^*, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta}) \big\} \subset \Theta. \tag{13}$$

Note that, in more complex problems, especially when it involves nonconvex optimization, we may get a local solution from (12) instead of the $\boldsymbol{\theta}_0$ that generated $\boldsymbol{y}_{obs}$. In this case, we refer to such a local solution $\boldsymbol{\theta}_0 = H(\boldsymbol{y}_{obs}, \boldsymbol{u}^{rel})$ as an *oracle parameter*. Our repro samples inference is then for this algorithm $H(\cdot,\cdot)$ specific parameter. Here, with slight abuse of notation, we still use $\boldsymbol{\theta}_0$ for the oracle parameter, and we further assume it is uniquely defined given the algorithm $H(\cdot,\cdot)$. The same model assumption (11) is also used in the development of *extended fiducial inference* (Liang et al., 2025), although the authors imposed a further requirement that $\boldsymbol{\theta}_0 = H(\boldsymbol{y}_{obs}, \boldsymbol{u}^{rel})$ is the model parameters $\boldsymbol{\theta}_0$ that generated $\boldsymbol{y}_{obs}$.

In this paper, we treat $g(\cdot,\cdot)$ or $H(\cdot,\cdot)$ as part of given model specification. The coverage results in Theorem 1 can be directly extended to the sets defined in (10) and (13), where the statements also cover the oracle parameter defined above. Due to space limit, the formal theorem, with its proof, is provided in Appendix A.2. Additionally, Appendix A.4 contains two illustrative examples of (10), where Example A1 demonstrates how to make a nonparametric inference for the quantile $\theta_0 = F^{-1}(\zeta)$ with a completely unknown distribution F and it shows that the repro samples method works well even for ζ near 0 or 1, whereas the conventional bootstrap method has issues on coverage. The method in Example A1 is also further utilized to provide a robust approach to estimating location and scale parameters in Gaussian mixture models. Further details are in Example A2 of Appendix A.4 and Section C.3.

Unless specified otherwise, in the rest of the paper, $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ refers to that in (3), instead of (10) or (13). All results developed for (3) have direct extensions for (10) and (13).

3 Further developments: repro samples method with mixed types of parameters

Inference problems in contemporary statistics often involves nuisance parameters and mixed types of parameters. In this section, we partition the parameters by $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}^\top)^\top$, for $\eta \in \Upsilon$ and $\boldsymbol{\beta} \in \mathcal{B}$. We start with Section 3.1 on a profile method for finite-sample inference for a general η. Then, we focus on the case of mixed types of parameters in Sections 3.2 and 3.3, where η is discrete or non-numeric and the remaining β may or may not depend on η. A practical guideline on use of repro samples methods is provided in Section 3.4.

3.1 A generic profile approach for finite inference

The generic profile method below does not need a plug-in estimator of nuisance parameter nor large-sample results; it guarantees finite-sample inference performance of target parameter.

Let $\boldsymbol{y} = G(\boldsymbol{\theta}, \boldsymbol{u})$ be a copy of sample data and $\Gamma_{1-\gamma}(\boldsymbol{y})$ be a level $1-\gamma$ confidence set for $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}^\top)^\top$. Let us consider another copy of parameter vector $\widetilde{\boldsymbol{\theta}} = (\eta, \widetilde{\boldsymbol{\beta}}^\top)^\top \in \Theta$ that has the same η but perhaps a different $\widetilde{\boldsymbol{\beta}} \neq \boldsymbol{\beta}$. To explore the impact of a potentially misspecified nuisance parameter $\widetilde{\boldsymbol{\beta}} \neq \boldsymbol{\beta}$, we define

$$\nu(\boldsymbol{u}; \eta, \boldsymbol{\beta}, \widetilde{\boldsymbol{\beta}}) = \inf\left\{1-\gamma : \widetilde{\boldsymbol{\theta}} \in \Gamma_{1-\gamma}(\boldsymbol{y})\right\} = \inf\left\{1-\gamma : \widetilde{\boldsymbol{\theta}} \in \Gamma_{1-\gamma}(G(\boldsymbol{\theta}, \boldsymbol{u}))\right\}. \quad (14)$$

From Corollary 1, we can interpret $1 - \nu(\boldsymbol{u}; \eta, \boldsymbol{\beta}, \widetilde{\boldsymbol{\beta}})$ as a 'p-value' when we test the null hypothesis that the data $\boldsymbol{y}$ is from $\widetilde{\boldsymbol{\theta}} = (\eta, \widetilde{\boldsymbol{\beta}}^\top)^\top$ but in fact $\boldsymbol{y} = G(\boldsymbol{\theta}, \boldsymbol{u})$ is from $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}^\top)^\top$.

To make inferences for the target η, we propose a *profile nuclear mapping function*:

$$T_p(\boldsymbol{u}, \boldsymbol{\theta}) = \min_{\widetilde{\boldsymbol{\beta}}} \ \nu\left(\boldsymbol{u}; \eta, \boldsymbol{\beta}, \widetilde{\boldsymbol{\beta}}\right). \quad (15)$$

Ideally, we would seek that the profile nuclear mapping $T_p(\boldsymbol{u}, \boldsymbol{\theta}) = T_p(\boldsymbol{u}, \eta)$ free of $\boldsymbol{\beta}$. However, even if $T_p(\boldsymbol{u}, \boldsymbol{\theta}) = T_p(\boldsymbol{u}, (\eta, \boldsymbol{\beta}^\top)^\top)$ still depends on the unknown nuisance $\boldsymbol{\beta}$, its random version $T_p(\boldsymbol{U}, \boldsymbol{\theta})$ is dominated by a $U(0,1)$ random variable as stated in Lemma 1.

Lemma 1. *Suppose $T_p(\boldsymbol{u}, \boldsymbol{\theta})$ is a profile nuclear mapping function from $\mathcal{U} \times \Theta \to [0,1]$*

defined in (15). Then, we have $\mathbb{P}\{T_p(\boldsymbol{U},\boldsymbol{\theta}) \le 1-\alpha\} \ge 1-\alpha$.

By Lemma 1, the Borel set for our repro samples method is $B_{1-\alpha} = (0, \alpha]$. We construct

$$\Xi_{1-\alpha}(\boldsymbol{y}_{obs}) = \left\{\eta : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ and } \boldsymbol{\beta}, \text{s.t.} \begin{pmatrix}\eta\\ \boldsymbol{\beta}\end{pmatrix} \in \Theta, \boldsymbol{y}_{obs} = G\left(\begin{pmatrix}\eta\\ \boldsymbol{\beta}\end{pmatrix}, \boldsymbol{u}^*\right), T_p\left(\boldsymbol{u}^*, \begin{pmatrix}\eta\\ \boldsymbol{\beta}\end{pmatrix}\right) \le 1-\alpha\right\}. \quad (16)$$

Theorem 3 below states that $\Xi_{1-\alpha}(\boldsymbol{y}_{obs})$ above is a level $1-\alpha$ confidence set for the target η.

Theorem 3. *Suppose* $\boldsymbol{Y} = G(\boldsymbol{\theta}, \boldsymbol{U})$ *with* $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, *where* $\boldsymbol{\theta}_0 = (\eta_0, \boldsymbol{\beta}_0^\top)^\top$ *is the true parameter, and* $\Xi_{1-\alpha}(\cdot)$ *is defined in* (16). *Then, we have* $\mathbb{P}\{\eta_0 \in \Xi_{1-\alpha}(\boldsymbol{Y})\} \ge 1-\alpha$.

In (16), we only collect η. If collecting both η and $\boldsymbol{\beta}$, we obtain a joint confidence set for $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}^\top)^\top$, but this joint set is unsuitable for joint inference because it imposes no constraints on the elements of $\boldsymbol{\beta}$. Nevertheless, since our interest is solely in η, we profile out the unconstrained $\boldsymbol{\beta}$ to obtain the confidence set of η in (16). In fact, we can obtain a confidence set for η from any joint confidence set of $\boldsymbol{\theta}$ by profiling out the nuisance $\boldsymbol{\beta}$. However, profiling out a set that places no restrictions on the nuisance parameter $\boldsymbol{\beta}$ is preferable to profiling a well-defined joint confidence set that does impose constraints on $\boldsymbol{\beta}$. For example, in a simple bivariate independent Gaussian case $(x_{i1}, x_{i2})^\top \sim N\big((\nu_1, \mu_2)^\top, \mathrm{diag}(1,1)\big)$, $i = 1, \ldots, n$, profiling out nuisance parameter μ_2 in the best 95% joint confidence set $\big\{(\mu_1, \mu_2) : (\mu_1 - \bar{x}_1)^2 + (\mu_2 - \bar{x}_2)^2 \le \frac{\chi^2_{2,0.95}}{n}\big\}$ leads to a 95% confidence interval of μ_1, $\big(\bar{x}_1 - \sqrt{\frac{\chi^2_{2,0.95}}{n}}, \bar{x}_1 + \sqrt{\frac{\chi^2_{2,0.95}}{n}}\big)$, which is worse than $\big(\bar{x}_1 - \frac{1.96}{\sqrt{n}}, \bar{x}_1 + \frac{1.96}{\sqrt{n}}\big)$, by profiling out the joint set $\{(\mu_1, \mu_2) : |\mu_1 - \bar{x}_1| \le \frac{1.96}{\sqrt{n}}\}$ with no constraint on μ_2. See also Michael et al. (2019) for a related discussion.

Finally, note that $T_p(\boldsymbol{u}, \boldsymbol{\theta})$ in (15) is defined through $\nu(\cdot; \eta, \boldsymbol{\beta}, \widetilde{\boldsymbol{\beta}})$ in (14), which often can be computed either directly or by a Monte-Carlo method. Due to space limit, we refer readers to Appendix B.2 and Lemma A1 for the Monte-Carlo method and its justification.

3.2 Fisher inversion and space reduction for discrete/non-numerical parameters

In this subsection, the target parameter $\eta \in \Upsilon$ is discrete or non-numerical, and $\boldsymbol{\beta}$ is nuisance parameters (of any types) that may even depend on η. We take advantage of an inherent "many-to-one" inversion mapping to get a data-dependent candidate set of η that is drastically smaller than Υ. By doing so, the discreteness of η, a hurdle for many large sample based inference methods, becomes an advantage for computation and theory derivation.

We first note that $\boldsymbol{y}_{obs} = G((\eta_0, \boldsymbol{\beta}_0^\top)^\top, \boldsymbol{u}^{rel})$, so we can rewrite η_0 as a solution of an optimization problem: $\eta_0 = \arg\min_\eta \min_{\boldsymbol{\beta}} \|\boldsymbol{y}_{obs} - G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^{rel})\|$ or, more generally,

$$\eta_0 = \arg\min_\eta \min_{\boldsymbol{\beta}} L\big(\boldsymbol{y}_{obs}, G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^{rel})\big), \tag{17}$$

where $L\big(\boldsymbol{y}, \boldsymbol{y}'\big)$ is a continuous loss function that measures the difference between two copies of data $\boldsymbol{y}$ and $\boldsymbol{y}'$. However, since we do not observe the unknown $\boldsymbol{u}^{rel}$, we replace $\boldsymbol{u}^{rel}$ with a repro copy $\boldsymbol{u}^*$ in (17). It leads to a function that maps a value $\boldsymbol{u}^* \in \mathcal{U}$ to a value $\eta^* \in \Upsilon$,

$$\eta^* = \arg\min_\eta \min_{\boldsymbol{\beta}} L\big(\boldsymbol{y}_{obs}, G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^*)\big). \tag{18}$$

We refer this operation as *Fisher inversion*, which can be traced back to Fisher's fiducial development (cf., Thornton and Xie, 2024). Here, $\mathcal{U}$ is often uncountable and Υ is countable, so (18) is "many-to-one": many $\boldsymbol{u}^* \in \mathcal{U}$ correspond to a single $\eta^* \in \Upsilon$. Of particularly interest is $S = \big\{\boldsymbol{u}^* : \eta_0 = \arg\min_\eta \min_{\boldsymbol{\beta}} L\big(\boldsymbol{y}_{obs}, G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^*)\big)\big\} \subset \mathcal{U}$, in which the mapping (18) produces $\eta^* = \eta_0$ for any $\boldsymbol{u}^* \in S$. Since $\boldsymbol{u}^{rel} \in S$, we know $S \neq \emptyset$.

If the subset $S \subset \mathcal{U}$ is a nontrivial set, i.e., $\mathbb{P}\big\{\boldsymbol{U} \in S \big| \boldsymbol{y}_{obs}\big\} > 0$ has a positive probability, we can use a Monte-Carlo method to get a candidate set $\widehat{\Upsilon} = \widehat{\Upsilon}(\boldsymbol{y}_{obs})$ such that $\eta_0 \in \widehat{\Upsilon}(\boldsymbol{y}_{obs})$ with a high (Monte-Carlo) probability. The idea is to simulate a sequence of $\boldsymbol{u}^s \sim \boldsymbol{U}$, say, $\mathcal{V} = \big\{\boldsymbol{u}_1^s, \ldots, \boldsymbol{u}_N^s\big\}$. If $|\mathcal{V}| = N$ is large enough, we have $\mathcal{V} \cap S \neq \emptyset$ and thus $\eta_0 \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$, with a high (Monte-Carlo) probability. Specifically, we define our candidate set as

$$\widehat{\Upsilon} = \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs}) = \big\{\eta^s = \arg\min_\eta \min_{\boldsymbol{\beta}} L\big(\boldsymbol{y}_{obs}, G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^s)\big) \;\big|\; \boldsymbol{u}^s \in \mathcal{V}\big\}. \tag{19}$$

Formally, let $\boldsymbol{U}^*$ be an independent copy of $\boldsymbol{U}$. We assume the following condition holds:

(N1) *For any $\boldsymbol{u}^{rel} \in \mathcal{U}$, there exists a neighborhood of $\boldsymbol{u}^{rel}$, say $S_{nb}(\boldsymbol{u}^{rel})$, such that*

$$S_{nb}(\boldsymbol{u}^{rel}) \subseteq S = \big\{\boldsymbol{u}^* : \eta_0 = \arg\min_\eta \min_\beta L\big(G((\eta_0, \boldsymbol{\beta}_0^\top)^\top, \boldsymbol{u}^{rel}), G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^*)\big)\big\},$$

and $\mathbb{P}_{\boldsymbol{U}}\left[\mathbb{P}_{\boldsymbol{U}^*|\boldsymbol{U}}\left\{\boldsymbol{U}^* \in S_{nb}(\boldsymbol{U}) | \boldsymbol{U}\right\} \geq c_{nb}\right] = 1$ *for a positive number $c_{nb} > 0$.*

Although the neighborhood S_{nb} and the number c_{nb} are problem specific, Condition (N1) is often satisfied when $L(\cdot, \cdot)$ is a continuous loss function and $\mathcal{U}$ is uncountable. Lemma 2 below states that, under Condition (N1), $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$ in (19) contains η_0 with high probability for a large $|\mathcal{V}|$. Here, $\mathbb{P}_{\boldsymbol{U},\mathcal{V}}(\cdot)$ refers to the joint distribution of $\boldsymbol{U}$ and Monte-Carlo $\boldsymbol{U}^*$'s in $\mathcal{V}$, and $\boldsymbol{Y} = G(\boldsymbol{\theta}, \boldsymbol{U})$ is generated with $\boldsymbol{\theta} = \boldsymbol{\theta}_0 = (\eta_0, \boldsymbol{\beta}_0^\top)^\top$.

Lemma 2. *If Condition (N1) holds, then* $\mathbb{P}_{\boldsymbol{U},\mathcal{V}}\big\{\eta_0 \notin \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})\big\} \leq (1 - c_{nb})^{|\mathcal{V}|}$.

As long as Condition (N1) holds, a neighborhood to recover the true model with (18) always exists. As a result, Lemma 2 and the following Theorem 4 always hold even in challenging cases where the dimension of η_0 increases with n. However, how small c_{nb} is depends on how many alternative η are close to η_0. In challenging cases such as when dimension of η_0 increases rapidly with n, or when weak signals are present in high-dimensional linear models, c_{nb} could be very small, and recovering η_0 with $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$ may demand a large $|\mathcal{V}|$.

In Theorem 4 below, the probability $\mathbb{P}_{\boldsymbol{U}|\mathcal{V}}(\cdot)$ refers to the condition distribution of $\boldsymbol{U}$ given the Monte-Carlo set $\mathcal{V}$ and the "in probability" statement is regarding to the Monte-Carlo randomness of $\mathcal{V}$. Theorem 4 has a clear frequentist statement: if we use the proposed method to obtain the candidate set $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$, we have a near 100% chance (a Monte-Carlo probability statement regarding the Monte-Carlo set $\mathcal{V}$) that $\eta_0 \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$ and in addition the frequency coverage probability of the repro sample confidence set $\Gamma'_{1-\alpha}(\boldsymbol{Y}) = \Gamma_{1-\alpha}(\boldsymbol{Y}) \cap \{\boldsymbol{\theta} : \eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})\}$ is at least $1-\alpha$ (the regular frequentist statement regarding $\boldsymbol{Y}$ or equivalently $\boldsymbol{U}$). Here again, $\boldsymbol{Y} = G(\boldsymbol{\theta}, \boldsymbol{U})$ is generated with $\boldsymbol{\theta} = \boldsymbol{\theta}_0 = (\eta_0, \boldsymbol{\beta}_0^\top)^\top$.

Theorem 4. *Suppose Condition (N1) holds and* $\widehat{\Upsilon}(\boldsymbol{y}_{obs})$ *is defined in* (19). *Then, (a) there exists a positive number c such that we have, in probability,* $\mathbb{P}_{\boldsymbol{U}|\mathcal{V}}\big\{\eta_0 \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})\big\} = 1 - o_p\big(e^{-c|\mathcal{V}|}\big)$. *(b) Let* $\Gamma'_{1-\alpha}(\boldsymbol{Y}) = \Gamma_{1-\alpha}(\boldsymbol{Y}) \cap \{\boldsymbol{\theta} : \eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})\}$ *where* $\Gamma_{1-\alpha}(\boldsymbol{Y})$ *is defined in Theorem 1 or A1, then* $\mathbb{P}_{\boldsymbol{U}|\mathcal{V}}\big\{\,\boldsymbol{\theta}_0 \in \Gamma'_{1-\alpha}(\boldsymbol{Y})\big\} \geq 1 - \alpha - o_p\left(e^{-c|\mathcal{V}|}\right)$, *in probability.*

In Theorem 4, the term $o_p\left(e^{-c|\mathcal{V}|}\right)$ tends to 0 as the size of the Monte-Carlo set $|\mathcal{V}| \to \infty$ and the results hold for any fixed (data) sample size $n < \infty$; so it is a finite-sample result. We will further discuss and elaborate Condition (N1) and Theorem 4 in Section 4.

Often times we can find an upper bound for the size of the candidate set $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$, because if η is far away from η_0, it usually has a small probability of entering $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$. Hence, $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$

usually only contains η that are relatively closer to η_0. The bound however is problem specific and depends on the loss function used in the inversion algorithm (18). In Theorem 7 of Section 4, we derive an upper bound for the candidate set of the unknown number of components in Gaussian mixture model. See also Theorem 3 of Wang et al. (2025) for a similar upper bound result on the size of a model candidate set for high-dimensional regressions.

3.3 A three-step procedure for inference with mixed-type parameters

In many applications, the target model parameters $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}^\top)^\top$ are mixed types and the parameter $\boldsymbol{\beta}$ may depend on η (e.g., Section 4). In this subsection, we propose a three-step inference procedure to dissect this difficult inference problem into manageable steps, in which we make inference for η, $\boldsymbol{\beta}$ and jointly $(\eta, \boldsymbol{\beta}^\top)^\top$, respectively. The procedure is as follows:

Step 1 [Inference for η_0] Use a repro samples method to construct a candidate set $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$ and then a level $1-\alpha$ confidence set $\Xi_{1-\alpha,1}(\boldsymbol{y}_{obs})$ for η_0, treating $\boldsymbol{\beta}$ as nuisance parameters.

Step 2: [Inference for $\boldsymbol{\beta}_0$] For a given $\eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$, use a repro samples method (or a regular inference approach) to construct a level $1-\alpha$ confidence set $\Xi_{1-\alpha,2}(\boldsymbol{y}_{obs})$ for $\boldsymbol{\beta}_0$.

Step 3: [Joint Inference for $(\eta_0, \boldsymbol{\beta}_0^\top)^\top$] Construct a level $1-\alpha$ joint confidence set, say $\Xi_{\alpha,3}(\boldsymbol{y}_{obs})$, for $\boldsymbol{\theta}_0 = (\eta_0, \boldsymbol{\beta}_0^\top)^\top$ based on the results in Steps 1 and 2.

The procedure breaks our task into three sub-tasks. In some situations, especially when $\boldsymbol{\beta}$ depends on η, it is necessary to handle η and $\boldsymbol{\beta}$ separately by steps. For instance, in the Gaussian mixture model in Section 4, $\eta = \tau$ is the unknown number of components, and the dimension of $\boldsymbol{\beta} = (\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$ is a function of τ — If $\tau = 3$, $\boldsymbol{\beta}$ has 6 parameters, while $\tau = 4$, $\boldsymbol{\beta}$ has 8 parameters; the μ_1 in $\tau = 3$ case may or may not line up with the μ_1 in $\tau = 4$ case. It is not clear to us how to make a meaningful inference for μ_1 without first working on τ.

In Step 1, we first use Section 3.2 and formula (19) to obtain a candidate set $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$. Then, we treat $\boldsymbol{\beta}$ as nuisance parameters and use Section 3.1 and formula (16) to construct:

$$\Xi_{1-\alpha,1}(\boldsymbol{y}_{obs}) = \left\{ \eta : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ and } \boldsymbol{\beta},\ \text{s.t.} \begin{pmatrix} \eta \\ \beta \end{pmatrix} \in \Theta, \boldsymbol{y}_{obs} = G\left(\begin{pmatrix} \eta \\ \beta \end{pmatrix}, \boldsymbol{u}^* \right), T_p\left(\boldsymbol{u}^*, \begin{pmatrix} \eta \\ \beta \end{pmatrix} \right) \le 1-\alpha \right\} \cap \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs}), \quad (20)$$

where $T_p\left(\boldsymbol{u}^*, (\eta, \boldsymbol{\beta})^\top\right)$ is defined following (15) for the discrete $\eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$.

In Step 2, for each $\eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$, we treat it as given so the only parameters now are $\boldsymbol{\beta}$. By using the repro samples approach in Section 2.1 (e.g., formula (3)) or using a regular inference approach, we can construct a *representative set* of $\boldsymbol{\beta}_0$, given a $\eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$:

$$\Gamma^{[\eta]}_{1-\alpha}(\boldsymbol{y}_{obs}) = \big\{\boldsymbol{\beta} : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } \boldsymbol{y}_{obs} = G^{[\eta]}(\boldsymbol{\beta}, \boldsymbol{u}^*),\, T^{[\eta]}(\boldsymbol{u}^*, \boldsymbol{\beta}) \in B^{[\eta]}_{1-\alpha}(\boldsymbol{\beta})\big\} \subset \mathcal{B}. \qquad (21)$$

Here, to highlight η is given, we denote $G^{[\eta]}(\boldsymbol{\beta}, \boldsymbol{u}^*) = G((\eta, \boldsymbol{\beta}^\top)^\top, \boldsymbol{u}^*)$, $T^{[\eta]}(\boldsymbol{u}^*, \boldsymbol{\beta}) = T(\boldsymbol{u}^*, (\eta, \boldsymbol{\beta}^\top)^\top)$ and $B^{[\eta]}_{1-\alpha}(\boldsymbol{\beta}) = B_{1-\alpha}((\eta, \boldsymbol{\beta}^\top)^\top)$. If $\eta = \eta_0$, $\Gamma^{[\eta]}_{1-\alpha}(\boldsymbol{y}_{obs})$ is a level $1-\alpha$ confidence set of $\boldsymbol{\beta}_0$. However, we do not know whether the given η is η_0, so we instead refer to $\Gamma^{[\eta]}_{1-\alpha}(\boldsymbol{y}_{obs})$ in (21) as a *representative set* of $\boldsymbol{\beta}_0$. To get a level $1-\alpha$ confidence set for $\boldsymbol{\beta}_0$ with an unknown η_0 and to accommodating the uncertainty of estimating η_0, we consider

$$\begin{aligned}\Xi_{1-\alpha,2}(\boldsymbol{y}_{obs}) &= \cup_{\boldsymbol{\eta}\in \widehat{\Upsilon}(\boldsymbol{y}_{obs})} \Gamma^{[\eta]}_{1-\alpha}(\boldsymbol{y}_{obs}) \\ &= \big\{\boldsymbol{\beta} : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } \boldsymbol{y}_{obs} = G^{[\eta]}(\boldsymbol{\beta}, \boldsymbol{u}^*),\, T^{[\eta]}(\boldsymbol{u}^*, \boldsymbol{\beta}) \in B^{[\eta]}_{1-\alpha}(\boldsymbol{\beta}), \eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})\big\}. \qquad (22)\end{aligned}$$

Alternatively, we may replace $\eta \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$ with $\eta \in \Xi_{1-\alpha',1}(\boldsymbol{y}_{obs})$, where $0 < \alpha' < \alpha$ and $\Xi_{1-\alpha,1}(\boldsymbol{y}_{obs})$ is a level-α' confidence set from η_0 obtained using Step 1. We then construct a representative set of $\boldsymbol{\beta}$, say $\Gamma^{[\eta]}_{1-\alpha''}(\boldsymbol{y}_{obs})$, as in (21) for $\alpha'' = \alpha - \alpha'$. Now, (22) is replaced by

$$\begin{aligned}\Xi_{1-\alpha,2}(\boldsymbol{y}_{obs}) &= \cup_{\boldsymbol{\eta}\in \Xi_{1-\alpha,1}(\boldsymbol{y}_{obs})} \Gamma^{[\eta]}_{1-\alpha''}(\boldsymbol{y}_{obs}) \\ &= \big\{\boldsymbol{\beta} : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } \boldsymbol{y}_{obs} = G^{[\eta]}(\boldsymbol{\beta}, \boldsymbol{u}^*),\, T^{[\eta]}(\boldsymbol{u}^*, \boldsymbol{\beta}) \in B^{[\eta]}_{\alpha''}(\boldsymbol{\beta}), \boldsymbol{\eta} \in \Xi_{1-\alpha',1}(\boldsymbol{y}_{obs})\big\}. \qquad (23)\end{aligned}$$

Since $\widehat{\Upsilon}(\boldsymbol{y}_{obs})$ can be viewed as a level 100% confidence set for η_0, (22) is a special case of (23) with $\alpha' = 1$ and $\alpha'' = \alpha$. Because $\boldsymbol{\beta}$'s length may vary for different η values, $\boldsymbol{\beta}$'s parameter space $\mathcal{B}$ (also the space $\Gamma^{[\eta]}_{1-\alpha''}(\boldsymbol{y}_{obs})$ is in) may have different dimensions for different η values. Thus, $\Xi_{\alpha,2}(\boldsymbol{y}_{obs})$ in (22) and (23) is possibly a union of sets of different dimensions.

In Step 3, the construction of a joint confidence set for $\boldsymbol{\theta}_0 = (\eta_0, \boldsymbol{\beta}_0^\top)^\top$ is similar to that of (23), but the confidence set is a subset in the space $\Theta = \Upsilon \otimes \mathcal{B}$, instead of $\mathcal{B}$; thus we use a product instead of a union to construct the joint confident set:

$$\begin{aligned}\Xi_{\alpha,3}(\boldsymbol{y}_{obs}) &= \Xi_{1-\alpha',1}(\boldsymbol{y}_{obs}) \otimes \Gamma^{[\eta]}_{1-\alpha''}(\boldsymbol{y}_{obs}) \qquad (24) \\ &= \big\{\boldsymbol{\theta} = (\eta, \boldsymbol{\beta}) : \exists\, \boldsymbol{u}^* \in \mathcal{U} \text{ s.t. } \boldsymbol{y}_{obs} = G^{[\eta]}(\boldsymbol{\beta}, \boldsymbol{u}^*),\, T^{[\eta]}(\boldsymbol{u}^*, \boldsymbol{\beta}) \in B^{[\eta]}_{\alpha''}(\boldsymbol{\beta}), \boldsymbol{\eta} \in \Xi_{1-\alpha',1}(\boldsymbol{y}_{obs})\big\}.\end{aligned}$$

Here, the product $\otimes$ refers to that an element in Υ and a (corresponding) element in $\mathcal{B}$ jointly form an element in Θ. Also, (α', α'') and other terms involved are defined as in (23).

Coverage statements of the above confidence sets are below, with a proof in Appendix B.1.

Corollary 3. *Under this subsection's setup and the conditions in Theorems 1 and 4, we have (a)* $\mathbb{P}\{\eta_0 \in \Xi_{1-\alpha,1}(\boldsymbol{Y})\} \geq 1 - \alpha - o_p(e^{-c_1|\mathcal{V}|})$; *(b)* $\mathbb{P}\{\boldsymbol{\beta}_0 \in \Xi_{1-\alpha,2}(\boldsymbol{Y})\} \geq 1 - \alpha - o_p(e^{-c_2|\mathcal{V}|})$; *(c)* $\mathbb{P}\{\boldsymbol{\theta}_0 \in \Xi_{\alpha,3}(\boldsymbol{Y})\} \geq 1 - \alpha - o_p(e^{-c_3|\mathcal{V}|})$. *Here,* $c_i > 0$ *is some constant for* $i = 1, 2, 3$.

We remark that the three-step procedure can be extended to the more general case of $\boldsymbol{\theta}_0 = (\eta_0, \boldsymbol{\beta}_0^\top, \boldsymbol{\varsigma}_0^\top)^\top$, where the inference targets are η_0, $\boldsymbol{\beta}_0$ and $(\eta_0, \boldsymbol{\beta}_0^\top)^\top$, and $\boldsymbol{\varsigma}_0$ are additional nuisance parameters. In this case, we need to slightly modify the formulas in (21) - (24), and still have results similar to Lemma 3. Due to space limit, the general formulas are not included in the paper; but we instead will consider some specific examples in Section 4. For instance, in Section 4.3 (also Section C.3), the target parameters are $(\tau_0, \mu_j^{(0)})$, for a given j, $1 \leq j \leq \tau_0$, where τ_0 is the unknown number of components and $\mu_j^{(0)}$ is the location parameter of the jth component in a Gaussian mixture model. In the example, $\eta_0 = \tau_0$, $\boldsymbol{\beta}_0 = \mu_j^{(0)}$ and $\boldsymbol{\varsigma}_0 = (\boldsymbol{M}_{\tau_0}, \boldsymbol{\mu}_{(-j)}^{(0)}, \boldsymbol{\sigma}_0)$, where $\boldsymbol{M}_{\tau_0}$ is the membership assignment matrix, $(\boldsymbol{\mu}_0, \boldsymbol{\sigma}_0)$ are the location and scale parameters of the τ_0 components, and $\boldsymbol{\mu}_{(-j)}^{(0)}$ is $\boldsymbol{\mu}_0$ minus $\mu_j^{(0)}$; see Section 4 for more discussions. Another example is in Wang et al. (2025) on the high-dimensional linear model, in which a repro samples method is used to construct a confidence set for a single coefficient, say $\boldsymbol{\beta}_j$, corresponding to covariate X_j. There, η_0 is the true sparse model, $\boldsymbol{\beta}_0 = \beta_j^{(0)}$ and $\boldsymbol{\varsigma}_0$ is the remaining $p - 1$ regression coefficients plus the error variance.

3.4 A practical guide and useful Monte-Carlo techniques

Based on the theory, the repro samples method is broadly applicable and covers most settings in which classical inference methods apply. In practice, it is particularly valuable in situations where the classical procedures are infeasible or perform poorly. In this subsection, we provide some guidelines and useful Monte-Carlo techniques for the practical use of the method.

To use a repro samples method, we often begin with a question: whether a classical inference approach is feasible and, if feasible, whether its performance is satisfactory. If the answer is no, the repro samples method could be a good alternative. As discussed earlier, most classical approaches rely on asymptotic theories with regularity conditions, and they may not perform well when the conditions are violated. Also, for discrete or non-numerical parameters and non-numerical data, inference approaches are limited. The repro samples method is most helpful in these cases.

In a classical inference approach, we often start with a probabilistic model, then identify a related test statistic and its distribution. Similarly, in our implementation, we also start with a model, either in a generative model form (1) or the more general setup of (8) or (9); we then identify a nuclear mapping $T(\boldsymbol{U}, \boldsymbol{\theta})$ for our inference problem and find a corresponding level $1-\alpha$ Borel set $B_{1-\alpha}(\boldsymbol{\theta})$. The choice of $T(\boldsymbol{U}, \boldsymbol{\theta})$ is similar to the choice of a test statistic in the classical approach that is problem specific. What $T(\boldsymbol{U}, \boldsymbol{\theta})$ we use may affect the efficiency and size of confidence set, but the validity of the confidence set is always guaranteed.

A very simple choice is $T(\boldsymbol{U}, \boldsymbol{\theta}) = T(\boldsymbol{U})$, free of $\boldsymbol{\theta}$, as in Examples 2 and A1 (Appenedix A.4); however, the suitability is problem dependent, and sometimes it may be inefficient or, in extreme cases, even can yield overly large and impractical confidence sets. Another common choice is $T(\boldsymbol{U}, \boldsymbol{\theta}) = \widehat{\boldsymbol{\theta}}(\boldsymbol{Y}_{\boldsymbol{\theta}})$, a point estimator of $\boldsymbol{\theta}$ using sample $\boldsymbol{Y}_{\boldsymbol{\theta}} = G(\boldsymbol{U}, \boldsymbol{\theta})$, hopefully an efficient estimator (for efficiency considerations). Moreover, in the case with nuisance parameters (Section 3.1), we may use $T(\boldsymbol{u}, \boldsymbol{\theta}) = \widehat{\eta}(\boldsymbol{Y}_{\boldsymbol{\theta}})$ to construct $\Gamma_{1-\gamma}(\boldsymbol{Y}_{\boldsymbol{\theta}})$ in (14) and obtain the profile nuclear mapping $T_p(\boldsymbol{u}, \boldsymbol{\theta})$ in (15). Here, and $\widehat{\eta}$ is an estimator of the target parameter η. Conditional methods may also be used to deal with nuisance parameters in some cases; cf., Wang et al. (2025). We may also explore other options when the point-estimator approach is undesirable; cf., Examples A4 and A3 of Appendix A.4.

After setting up the model and nuclear mapping, a generic implementation algorithm is:

When the distribution of the nuclear mapping $T(\boldsymbol{U}, \boldsymbol{\theta})$ is explicit, the confidence set $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ often also has an explicit form; cf., e.g., Examples 1, 2, A4 and A1. Under more complex

Algorithm 1 A generic implementation algorithm

1: For a given value of $\boldsymbol{\theta} \in \Theta$, (a) Obtain a level $1-\alpha$ Borel set $B_{1-\alpha}(\boldsymbol{\theta})$ in (2); (b) Check whether there exists a $\boldsymbol{u}^* \in \mathcal{U}$ such that $\boldsymbol{y}_{obs} = G(\boldsymbol{\theta}, \boldsymbol{u}^*)$ and $T(\boldsymbol{u}^*, \boldsymbol{\theta}) \in B_{1-\alpha}(\boldsymbol{\theta})$. If both of the above criteria are satisfied, keep the $\boldsymbol{\theta}$.

2: Collect all kept $\boldsymbol{\theta}$ to form a level $1-\alpha$ confidence set $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$.

settings with the distribution of $T(\boldsymbol{U}, \boldsymbol{\theta})$ not explicitly available, we may need to use Monte-Carlo methods. For instance, in Step 1(a), we can typically collect many simulated copies of $\boldsymbol{u}^s \sim \boldsymbol{U}$ to form a set $\mathcal{V}$. We then compute $\{T(\boldsymbol{u}^s, \boldsymbol{\theta}), \boldsymbol{u}^s \in \mathcal{V}\}$, for a given $\boldsymbol{\theta} \in \Theta$ and derive an empirical distribution of $T(\boldsymbol{U}, \boldsymbol{\theta})$, based on which we can obtain a level $1-\alpha$ Borel set $B_{1-\alpha}(\boldsymbol{\theta})$. In particular, if $T(\boldsymbol{u}, \boldsymbol{\theta})$ is a scalar, we can directly use its (upper/lower) quantiles of to get a level $1-\alpha$ interval as the Borel set $B_{1-\alpha}(\boldsymbol{\theta})$. When $T(\boldsymbol{u}, \boldsymbol{\theta})$ is a vector in $\mathcal{T} \subseteq \mathbb{R}^q$, we may use the data depth approach and follow Liu et al. (2022, §3.2.2) to construct the Borel set $B_{1-\alpha}(\boldsymbol{\theta})$; see further details in Appendix B.3.1. Alternatively, we may also get an (empirical) central region $B_{1-\alpha}(\boldsymbol{\theta})$ on $\mathcal{T}$ using *empirical center-outward quantiles* (developed based on optimal transport mapping); cf., Hallin et al. (2021) for further details.

When the distribution of $T(\boldsymbol{U}, \boldsymbol{\theta})$ depends on $\boldsymbol{\theta}$ and the corresponding $\Gamma_{1-\alpha}(\boldsymbol{y}_{obs})$ does not have an explicit form, Algorithm 1 needs to search over $\boldsymbol{\theta} \in \Theta$, which can be computationally expensive. Fortunately, in most situations, we can avoid such an exhaustive search over $\boldsymbol{\theta}$. In this paper, we highlight two techniques that can help mitigate this potential computational issue: one is the previously discussed technique in Section 3.2 for discrete or non-numerical parameters where candidate sets are used to reduce search spaces, and the other is for continuous parameters using a quantile regression approach implemented in Dalmasso et al. (2024). In the case of mixed types of parameters, the two techniques can be used together.

Specifically, on the quantile-regression technique, suppose our target parameter $\boldsymbol{\theta}$ is in a continuous space Θ and that $T(\cdot, \boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta}$. We use the following technique to improve our computation efficiency: *Let $\pi(\boldsymbol{\theta})$ be a proposal distribution that has no-zero probability support on Θ. In addition to $|\mathcal{V}|$ copies of $\boldsymbol{u}^s \sim \boldsymbol{U}$, we simulate $|\mathcal{V}|$ copies of $\boldsymbol{\theta}^s \sim$*

$\pi(\boldsymbol{\theta})$. *We then fit a level* $1-\alpha$ *nonparametric quantile regression model with* $t^s = T(\boldsymbol{u}^s, \boldsymbol{\theta}^s)$ *as response and* $\boldsymbol{\theta}^s$ *as covariates to output the level* $1-\alpha$ *conditional quantile curve, say* $\widehat{C}_{1-\alpha}(\theta)$*. The Borel set* $B_{1-\alpha}(\boldsymbol{\theta})$ *for our repro samples method is either* $\big(-\infty, \widehat{C}_{1-\alpha}(\boldsymbol{\theta})\big)$ *or* $\big(\widehat{C}_{\frac{\alpha}{2}}(\boldsymbol{\theta}), \widehat{C}_{1-\frac{\alpha}{2}}(\boldsymbol{\theta})\big)$*.* This method is similar to Algorithm 1 of Dalmasso et al. (2024) where the authors use a quantile regression to get a level α rejection region in a Neyman test problem. The difference is that the t^s in Dalmasso et al. (2024) is computed from a test statistic, while our t^s comes from a nuclear mapping. Therefore we need not generate any data points $\boldsymbol{y}^s = G(\boldsymbol{\theta}^s, \boldsymbol{u}^s)$, making it simpler, especially when generating $\boldsymbol{y}^s$ is computationally expensive. See Appendix A.3 for an illustration of the method for Example 1. Dalmasso et al. (2024, Theorems 1) show that, as $|\mathcal{V}| \to \infty$, $\widehat{C}_q(\boldsymbol{\theta})$ is the conditional level q quantile, if the quantile regression estimator is consistent. Similarly, in our case, if the quantile regression estimator is consistent, then $B_{1-\alpha}(\boldsymbol{\theta}) = \big(-\infty, \widehat{C}_{1-\alpha}(\theta)\big)$ or $B_{1-\alpha}(\boldsymbol{\theta}) = \big(\widehat{C}_{\frac{\alpha}{2}}(\boldsymbol{\theta}), \widehat{C}_{1-\frac{\alpha}{2}}(\boldsymbol{\theta})\big)$ satisfies (2). Furthermore, the machine learning technique to learn a test statistic in *Simulation Based Inference* (SBI) (Dalmasso et al., 2024; Tomaselli et al., 2025) may also be adopted to learn nuclear mapping $T(\boldsymbol{U}, \boldsymbol{\theta})$; cf., Tomaselli et al. (2025) for further details on SBI. More detailed discussion of the quantile regression technique can further be found in Appendix B.3.2.

When $\boldsymbol{\theta}$ is discrete or non-numerical, we can directly apply the technique in Section 3.2 to significantly reduce the search space of $\boldsymbol{\theta}$. More commonly when $\boldsymbol{\theta} = (\eta, \boldsymbol{\beta})$ is mixed types, the three-step approach in Section 3.3 is effective. To save computing costs, in Step 1 we can use the technique from Section 3.2, and in Steps 2-3, for a given η, we may use the quantile regression technique described above, especially when the representative set of $\boldsymbol{\beta}$ does not have an explicit form. See Sections 4 and 5 below for some implementation details.

4 Gaussian mixture model with unknown number of components

The Gaussian mixture model has a long history, and frequently used. However, making inference for the unknown number of mixture components, say τ_0, remains a "highly non-trivial" problem (cf., Wasserman et al., 2020). Although there exist several procedures to obtain point estimators of τ_0, assessing the uncertainty associated with these estimations remains largely open. This difficulty stems from the fact that the parameter τ_0 is a discrete integer and the standard tools relying on CLT are no longer applicable. Moreover, the uncertainty in τ_0 propagates to the inference on other model parameters, whose dimension and structure depend on the unknown τ_0. This section addressed these challenges by using the repro samples method to construct a confidence set for τ_0, and we also provide inference solutions for other parameters while accommodating the uncertainty in the unknown τ_0.

Consider a τ-component Gaussian mixture model: $Y_i, i = 1, \ldots, n$, is drawn from one of the component distributions $\{N(\mu_k, \sigma_k^2) : k = 1, \ldots, \tau\}$. Let $M_\tau = (m_{ik}) \in \{0, 1\}^{n\times\tau}$ be the (unobserved) membership matrix, with $m_{ik} = 1$ if $Y_i \sim N(\mu_k, \sigma_k^2)$ and $m_{ik} = 0$ otherwise. Then, the model parameter is $\boldsymbol{\theta} = (\tau, \boldsymbol{M}_\tau, \boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau^2)^\top$, where $\boldsymbol{\mu}_\tau = (\mu_j)_{\tau\times 1}$ and $\boldsymbol{\sigma}_\tau^2 = (\sigma_j^2)_{\tau\times 1}$. And the true values $\boldsymbol{\theta}_0 = (\tau_0, \boldsymbol{M}_0, \boldsymbol{\mu}_0, \boldsymbol{\sigma}_0^2)^\top$, where $\boldsymbol{\mu}_0 = (\mu_j^{(0)})_{\tau\times 1}, \boldsymbol{\sigma}_0^2 = (\sigma_j^{(0)^2})_{\tau\times 1}$.

We can re-express Y_i as a generative model $Y_i = \sum_{k=1}^{\tau} m_{ik}(\mu_k + \sigma_k U_i)$, $U_i \sim N(0, 1)$. In a vector form, $\boldsymbol{Y} = \boldsymbol{M}\boldsymbol{\mu} + \mathrm{diag}(\boldsymbol{U})\boldsymbol{M}\boldsymbol{\sigma} = \boldsymbol{M}\boldsymbol{\mu} + \mathrm{diag}(\boldsymbol{M}\boldsymbol{\sigma})\boldsymbol{U}$, where $\boldsymbol{Y} = (Y_1, \ldots, Y_n)^\top$ and $\boldsymbol{U} = (U_1, \ldots, U_n)^\top$. The realization of $\boldsymbol{Y}$ with true $\boldsymbol{\theta}_0$ is $\boldsymbol{y}_{obs} = \boldsymbol{M}_0\boldsymbol{\mu}_0 +\ \mathrm{diag}(\boldsymbol{u}^{rel})\boldsymbol{M}_0\boldsymbol{\sigma}_0$. An alternative setup is $Y_i \sim \sum_{k=1}^{\tau} p_k N(\mu_k, \sigma_k^2)$, which further assumes that $\boldsymbol{m}_i = (m_{i1}, \ldots, m_{i\tau})^\top$ is a random draw from multinomial distribution Multinomial$(\tau; p_1, \ldots, p_\tau)$ with unknown proportion parameters $(p_1, \ldots, p_\tau)$, $\sum_{k=1}^{\tau} p_k = 1$ (Chen et al., 2012). In this case, $\boldsymbol{M}_0$ is the realized membership assignment matrix that generates $\boldsymbol{y}_{obs}$.

It is known in the literature that there is an identifiability issue in a mixture model; i.e., multiple sets of parameters could possibly generate the same data (Liu and Shao, 2003). To account for this issue, we re-define the true parameters as the ones with the smallest τ, i.e.,

$$(\tau_0, \boldsymbol{M}_0, \boldsymbol{\mu}_0, \boldsymbol{\sigma}_0^2) = \arg\min_{\{(\tau, \boldsymbol{M}_\tau, \boldsymbol{\mu}_\tau, \boldsymbol{\sigma}^2) | \boldsymbol{M}_\tau\boldsymbol{\mu}_\tau = \boldsymbol{M}_0\boldsymbol{\mu}_0,\, \boldsymbol{M}_\tau\boldsymbol{\sigma}_\tau = \boldsymbol{M}_0\boldsymbol{\sigma}_0\}} \tau. \tag{25}$$

Here, $\boldsymbol{M}_0$ and $\boldsymbol{\mu}_0$ on the right-hand side are the true parameter values that generate $\boldsymbol{y}^{obs}$. With a slight abuse of notation (but for simplicity), we still use the same notations on the left-hand side to denote the ones with the smallest τ. We further assume that $(\tau_0, \boldsymbol{M}_0, \boldsymbol{\mu}_0, \boldsymbol{\sigma}_0^2)$ defined in (25) is unique and $0 < \sigma_{\max} = \|\boldsymbol{\sigma}_0\|_\infty < \infty$. Thus, among the τ_0 components, no σ_k^2 can completely dominate another σ_j^2, for any $j \neq k$. Moreover, we follow the existing literature to assume that the smallest number of components in (25) $\tau_0 = O(\log(n))$, since increasing τ_0 beyond $O(\log(n))$ only delivers indistinguishable gains in the likelihood function and one can confine attention to mixing distributions with no more than $O(\log(n))$ components; cf., Ghosal and van der Vaart (2001) and Wu and Polyanskiy (2025). The inference procedure we develop in this paper is for the set of parameters defined in (25). Due to space limit, we only include in the main text our inference for τ_0, with the detailed discussion for the location and scale parameters $\{(\mu_j^{(0)}, \sigma_j^{(0)}) : j = 1, \ldots, \tau_0\}$ placed in Appendix C.3.

4.1 A confidence set for the unknown number of components τ_0

When we work on τ, the other $(\boldsymbol{M}_\tau, \boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$ are nuisance parameters whose dimensions are functions of τ. Given $(\tau, \boldsymbol{M}_\tau)$, we can obtain sufficient statistics for $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau^2)$ by projecting $\boldsymbol{Y}$ onto the space expanded by $\boldsymbol{M}_\tau$. Thus, $T(\boldsymbol{U}, \boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{Y}, (\tau, \boldsymbol{M}_\tau))$ is free of $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$. We first use the profiling method in Section 3.1 (setting $\eta = \tau$ and $\boldsymbol{\beta} = \boldsymbol{M}_\tau$) to handle nuisance parameter $\boldsymbol{M}_\tau$, and then use a conditional method to address nuisance parameter $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$.

Assume, temporarily, we can obtain a nuclear mapping $T(\boldsymbol{U}, \boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{Y}, (\tau, \boldsymbol{M}_\tau))$ and have a corresponding Borel set $B_{1-\alpha}$ that satisfies $\mathbb{P}(T(\boldsymbol{U}, \boldsymbol{\theta}) \in B_{1-\alpha}) \geq 1-\alpha$. Then, following Section 3.1, for $\boldsymbol{y}$ that is generated from given parameters $(\tau, \boldsymbol{M}_\tau)$, we test whether it is from $(\tau, \widetilde{\boldsymbol{M}_\tau})$, $\widetilde{\boldsymbol{M}_\tau} \neq \boldsymbol{M}_\tau$. As in (14) and (15), we define a profile nuclear mapping for τ:

$$T_p(\boldsymbol{u}, \boldsymbol{\theta}) = \widetilde{T}_p(\boldsymbol{y}, \tau) = \min_{\widetilde{\boldsymbol{M}_\tau}} \widetilde{\nu}(\boldsymbol{y}, \tau, \widetilde{\boldsymbol{M}_\tau}), \tag{26}$$

where $\nu(\boldsymbol{u}, \tau, \widetilde{\boldsymbol{M}_\tau}) = \widetilde{\nu}(\boldsymbol{y}, \tau, \widetilde{\boldsymbol{M}_\tau}) = \inf\{1-\gamma : \widetilde{T}(\boldsymbol{y}, (\tau, \widetilde{\boldsymbol{M}_\tau})) \in B_{1-\gamma}\}$. By Lemma 1 and (16),

$$\Xi_{1-\alpha}(\boldsymbol{y}_{obs}) = \big\{\tau : \widetilde{T}_p(\boldsymbol{y}_{obs}, \tau) \leq 1 - \alpha\big\}; \tag{27}$$

detailed steps of deriving (27) using (16) are provided in Appendix C.1.1. Corollary 4 below states that $\Xi_{1-\alpha}(\boldsymbol{y}_{obs})$ in (27) is a level $1 - \alpha$ confidence set; A proof is in Appendix C.1.2.

Corollary 4. *Let $\boldsymbol{y} = \boldsymbol{M}_\tau \boldsymbol{\mu} + diag(\boldsymbol{M}_\tau \boldsymbol{\sigma})\boldsymbol{u}$, for fixed $\boldsymbol{\theta} = (\tau, \boldsymbol{M}_\tau, \boldsymbol{\mu}, \boldsymbol{\sigma})$, and $T(\boldsymbol{U}, \boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{y}, (\tau, \boldsymbol{M}_\tau))$. Suppose $\mathbb{P}(T(\boldsymbol{U}, \boldsymbol{\theta}) \in B_{1-\alpha}) \geq 1 - \alpha$ for a Borel set $B_{1-\alpha}$, given $\boldsymbol{\theta}$. If $\widetilde{T}_p(\boldsymbol{y}, \tau)$*

is defined by (26) *and* $\Xi_{1-\alpha}(\boldsymbol{y}_{obs})$ *is defined by* (27)*, then* $\mathbb{P}(\tau_0 \in \Xi_{1-\alpha}(\boldsymbol{Y})) \geq 1-\alpha$ *for a random sample* $\boldsymbol{Y}$ *that is generated with the set* $\boldsymbol{\theta} = \boldsymbol{\theta}_0 = (\tau_0, \boldsymbol{M}_0, \boldsymbol{\mu}_0, \boldsymbol{\sigma}_0)$*.*

We now move on to find $T(\boldsymbol{U}, \boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{y}, (\tau, \boldsymbol{M}_\tau))$ that satisfies the requirement in Corollary 4. To do so, let $\hat{\tau}(\boldsymbol{y})$ be an estimator of τ; in our paper we a BIC-type estimator:

$$\begin{aligned}\hat{\tau}(\boldsymbol{y}) &= \arg\min_{\tau\in\mathcal{T}} \left[\left\{-2\max\nolimits_{(\tau,\boldsymbol{M}_\tau,\boldsymbol{\mu},\boldsymbol{\sigma})}\log\ell(\boldsymbol{M}_\tau,\boldsymbol{\mu},\boldsymbol{\sigma}|\boldsymbol{y})\right\} + 2\tau\log(n)\right], \\ &= \arg\min_{\tau\in\mathcal{T}}\min_{\boldsymbol{M}_\tau}\left\{2\sum\nolimits_{k=1}^{\tau} n_k\log\left(\|(I-\boldsymbol{H}_\tau^{(k)})\boldsymbol{y}^{(k)}\|\right) - \sum\nolimits_{k=1}^{\tau} n_k\log n_k + 2\tau\log(n)\right\}, \quad (28)\end{aligned}$$

where $\ell(\tau, \boldsymbol{M}_\tau, \boldsymbol{\mu}, \boldsymbol{\sigma}|\boldsymbol{y}) = \sum_{k=1}^{\tau}\{-n_k\log(\sigma^{(k)}) - \frac{1}{2(\sigma^{(k)})^2}\|\boldsymbol{y}^{(k)} - \mu^{(k)}\|^2\}$ is the log-likelihood function of $(\tau, \boldsymbol{M}_\tau, \boldsymbol{\mu}, \boldsymbol{\sigma})$ given $\boldsymbol{y}$, $\boldsymbol{H}_\tau^{(k)}$=$\boldsymbol{H}_\tau[\mathcal{I}_k, \mathcal{I}_k]$ is the submatrix of $\boldsymbol{H}_\tau = \boldsymbol{M}_\tau(\boldsymbol{M}_\tau^\top\boldsymbol{M}_\tau)^{-1}\boldsymbol{M}_\tau^\top$ corresponding to the kth component, and $\boldsymbol{y}^{(k)} = \boldsymbol{y}[\mathcal{I}_k]$ is the corresponding subvector of $\boldsymbol{y}$. Here, $\mathcal{I}_k = \{i : m_{ik} = 1\}$ is the index set for component k and, following the common practice in the literature (cf., Chen, 2017; Fraley and Raftery, 1998, 2002), we set $\mathcal{T} = \{1, \ldots, K_{\max}\}$ for a fixed $K_{\max}$. In the next paragraph, we define our $T(\boldsymbol{U}, \boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{y}, (\tau, \boldsymbol{M}_\tau))$ via this point estimator $\hat{\tau}(\boldsymbol{Y})$ given $(\tau, \boldsymbol{M}_\tau)$, while accommodating the unknown $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$.

Suppose for the moment that we can compute $P_{\boldsymbol{Y}}(\bar{\tau}) = \mathbb{P}_{\boldsymbol{Y}}\{\hat{\tau}(\boldsymbol{Y}) = \bar{\tau}\}$, for any $\bar{\tau} \in \mathcal{T}$, where $\mathbb{P}_{\boldsymbol{Y}}\{\cdot\}$ refers the probability of the random sample $\boldsymbol{Y}$ that is generated using $\boldsymbol{U}$ and the set of fixed $\boldsymbol{\theta} = (\tau, \boldsymbol{M}_\tau, \boldsymbol{\mu}, \boldsymbol{\sigma})$. Then, following the practice of defining a p-value, we assess how 'extreme' the realized $\hat{\tau}(\boldsymbol{y})$ is, i.e., $\mathcal{P}(\boldsymbol{y}) = \sum_{\{\bar{\tau}:\, P_{\boldsymbol{Y}}(\bar{\tau}) > P_{\boldsymbol{Y}}(\hat{\tau}(\boldsymbol{y}))\}} P_{\boldsymbol{Y}}(\bar{\tau})$. We expect the distribution of $\mathcal{P}(\boldsymbol{Y})$ is closely related to $U(0,1)$. However, $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$ values are unknown, so we cannot directly compute the distribution $P_{\boldsymbol{Y}}(\bar{\tau})$, nor simulate $\boldsymbol{Y} = \boldsymbol{M}_\tau\boldsymbol{\mu}_\tau + \mathrm{diag}(\boldsymbol{M}_\tau\boldsymbol{\sigma}_\tau)\boldsymbol{U}$, when we are only given $(\tau, \boldsymbol{M}_\tau)$. Nevertheless, when $(\tau, \boldsymbol{M}_\tau)$ is given, $\big(\mathbf{a}_\tau(\boldsymbol{y}),\ \mathbf{b}_\tau(\boldsymbol{y})\big)$ are sufficient statistics for the unknown $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$, where $\mathbf{a}_\tau(\boldsymbol{y}) = (\boldsymbol{M}_\tau^\top\boldsymbol{M}_\tau)^{-1}\boldsymbol{M}_\tau^\top\boldsymbol{y}$ and $\mathbf{b}_\tau(\boldsymbol{y}) = \big(\|(I-\boldsymbol{H}_\tau^{(1)})\boldsymbol{y}^{(1)}\|, \cdots, \|(I-\boldsymbol{H}_\tau^{(\tau)})\boldsymbol{y}^{(\tau)}\|\big)^T$ are the "point estimators" of $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$ with observing sample data $\boldsymbol{y}$ and given $(\tau, \boldsymbol{M}_\tau)$. Thus, we can generate a repro sample $\boldsymbol{Y}'$ using $(\tau, M_\tau, \mathbf{a}_\tau(\boldsymbol{y}), \mathbf{b}_\tau(\boldsymbol{y}))$ instead. Specifically, we simulate an independent copy $\boldsymbol{U}' \sim \boldsymbol{U}$ and let $\boldsymbol{Y}' = \boldsymbol{M}_\tau\mathbf{a}_\tau(\boldsymbol{y}) + \mathrm{diag}\{\boldsymbol{M}_\tau\mathbf{b}_\tau(\boldsymbol{y})\}\,\mathbf{c}_\tau(\boldsymbol{U}')$, where $\mathbf{c}_\tau(\boldsymbol{U}') = \big(\frac{\{(I-\boldsymbol{H}_\tau^{(1)})\boldsymbol{U}'^{(1)}\}^T}{\|(I-\boldsymbol{H}_\tau^{(1)})\boldsymbol{U}'^{(1)}\|}, \cdots, \frac{\{(I-\boldsymbol{H}_\tau^{(\tau)})\boldsymbol{U}'^{(\tau)}\}^T}{\|(I-\boldsymbol{H}_\tau^{(\tau)})\boldsymbol{U}'^{(\tau)}\|}\big)^T$ and $\boldsymbol{U}'^{(k)} = \boldsymbol{U}'[\mathcal{I}_k]$. We then define our mapping as

$$T(\boldsymbol{U},\boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{y},(\tau,\boldsymbol{M}_\tau)) = \sum\nolimits_{\{\bar\tau: P_{\boldsymbol{Y}'}(\bar\tau) > P_{\boldsymbol{Y}'}(\hat\tau(\boldsymbol{y}))\}} P_{\boldsymbol{Y}'}(\bar\tau). \tag{29}$$

Since $\mathbf{c}_\tau(\boldsymbol{U}') \sim \mathbf{c}_\tau(\boldsymbol{U})$ and $\mathbf{c}_\tau(\boldsymbol{U})$ is independent of $(\mathbf{a}_\tau(\boldsymbol{Y}),\mathbf{b}_\tau(\boldsymbol{Y}))$, the repro samples $\boldsymbol{Y}'$ and $\boldsymbol{Y}$ have the same conditional distribution, i.e., $\boldsymbol{Y}'\big|\{\mathbf{a}_\tau(\boldsymbol{y}),\mathbf{b}_\tau(\boldsymbol{y})\} \sim\ \boldsymbol{Y}\big|\{\mathbf{a}_\tau(\boldsymbol{Y})\mathbf{b}_\tau(\boldsymbol{Y})\} = \{\mathbf{a}_\tau(\boldsymbol{y}),\mathbf{b}_\tau(\boldsymbol{y})\}$. Based on this fact, we have the theorem below; a proof is in Appendix C.1.2.

Theorem 5. *Given $\boldsymbol{\theta}$ and $T(\boldsymbol{U},\boldsymbol{\theta})$ in (29), we have $\mathbb{P}(T(\boldsymbol{U},\boldsymbol{\theta}) \le 1-\alpha) \ge 1-\alpha$.*

By Theorem 5 and (29), $\mathbb{P}(\widetilde{T}(\boldsymbol{Y},(\tau,\boldsymbol{M}_\tau)) \in B_{1-\alpha}) \ge 1-\alpha$, for $B_{1-\alpha} = [0, 1-\alpha]$. Thus, the function $T_p(\boldsymbol{u},\boldsymbol{\theta})$ in (26) can be simplified to be $T_p(\boldsymbol{u},\boldsymbol{\theta}) = \min_{\widetilde{\boldsymbol{M}}_\tau} \inf\{1-\alpha' : \widetilde{T}(\boldsymbol{y},(\tau,\widetilde{\boldsymbol{M}_\tau})) \le 1-\alpha'\} = \min_{\widetilde{\boldsymbol{M}}_\tau} \widetilde{T}(\boldsymbol{y},\, (\tau,\widetilde{\boldsymbol{M}_\tau}))$. Therefore, the confidence set in (27) becomes

$$\Xi_{1-\alpha}(\boldsymbol{y}_{obs}) = \left\{\tau : \min\nolimits_{\boldsymbol{M}_\tau} \widetilde{T}(\boldsymbol{y}_{obs},(\tau,\boldsymbol{M}_\tau)) \le 1-\alpha\right\}. \tag{30}$$

To obtain (30), it suffices to compute $\widetilde{T}(\boldsymbol{y}_{obs},(\tau,\boldsymbol{M}_\tau))$ for any given $(\tau,\boldsymbol{M}_\tau)$. We use a Monte-Carlo method to compute $\widetilde{T}(\boldsymbol{y}_{obs},(\tau,\boldsymbol{M}_\tau))$. Specifically, we simulate many $\boldsymbol{u}^s \sim \boldsymbol{U}$ and collect them to form a set $\mathcal{V}$. For each $\boldsymbol{u}^s \in \mathcal{V}$ and given $(\tau,\boldsymbol{M}_\tau)$, we compute $\boldsymbol{y}^s = \boldsymbol{M}_\tau\mathbf{a}_\tau(\boldsymbol{y}_{obs}) + \mathrm{diag}\{\boldsymbol{M}_\tau\mathbf{b}_\tau(\boldsymbol{y}_{obs})\}\mathbf{c}_\tau(\boldsymbol{u}^s)$ and $\hat\tau(\boldsymbol{y}^s)$. Then $\widehat{P}_{\mathcal{V}}(\bar\tau) = \frac{1}{|\mathcal{V}|}\sum_{\boldsymbol{u}^s\in\mathcal{V}} I(\hat\tau(\boldsymbol{y}^s) = \bar\tau)$ is an approximation of $P_{\boldsymbol{Y}'}(\bar\tau)$ for $\boldsymbol{Y}' = \boldsymbol{M}_\tau\mathbf{a}_\tau(\boldsymbol{y}_{obs}) + \mathrm{diag}\left\{\boldsymbol{M}_\tau\mathbf{b}_\tau(\boldsymbol{y}_{obs})\right\}\mathbf{c}_\tau(\boldsymbol{U}')$, $\boldsymbol{U}' \sim \boldsymbol{U}$. When $|\mathcal{V}|$ is large enough, we can approximate $\widetilde{T}(\boldsymbol{y}_{obs},(\tau,\boldsymbol{M}_\tau)) \approx \sum_{\{\bar\tau:\hat P_{\mathcal{V}}(\bar\tau) > \hat P_{\mathcal{V}}(\hat\tau(\boldsymbol{y}_{obs}))\}} \hat P_{\mathcal{V}}(\bar\tau)$.

4.2 A candidate set of $(\tau_0, \boldsymbol{M}_0)$ and a modified confidence set for τ_0

To avoid searching through all values of $(\tau,\boldsymbol{M}_\tau)$ in (30), we follow Section 3.2 to construct a data-driven candidate set $\widehat{\Upsilon}(\boldsymbol{y}_{obs})$. In the notation of Section 3.2, the target parameter for the development is $\eta = (\tau,\boldsymbol{M}_\tau)$. Corresponding to (18), we use a modified BIC as our lost function and define our many-to-one mapping for $\eta^* = (\tau^*,\boldsymbol{M}_\tau^*)$. Specifically, we define

$$(\tau^*,\boldsymbol{M}_{\tau^*}^*) = \arg\min_{(\tau,\boldsymbol{M}_\tau)}\min_{(\boldsymbol{\mu}_\tau,\bar\sigma)} \left\{n\log\left(\tfrac{\|\boldsymbol{y}_{obs}-\boldsymbol{M}_\tau\boldsymbol{\mu}_\tau-\bar\sigma\boldsymbol{u}^*\|^2+1}{n}\right) + 2\lambda\tau\log(n)\right\} \tag{31}$$

with a tuning $\lambda > 0$, where we only use a scalar $\bar\sigma$ instead of a vector $\boldsymbol{\sigma}_\tau$ (which significantly reduces the complexities of computing and theoretical developments). Due to space limit, the rational of using (31) is provided in Appendix C.2. By (19), a candidate set for $(\tau_0,\boldsymbol{M}_0)$ is

$$\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs}) = \left\{(\tau^*,\boldsymbol{M}_{\tau^*}^*) = \arg\min_{(\tau,\boldsymbol{M}_\tau)}\min_{(\boldsymbol{\mu}_\tau,\bar\sigma)} \left\{n\log\left(\tfrac{\|\boldsymbol{y}_{obs}-\boldsymbol{M}_\tau\boldsymbol{\mu}_\tau-\bar\sigma\boldsymbol{u}^s\|^2+1}{n}\right) + \lambda\log(n)(2\tau)\right\}, \boldsymbol{u}^s \in \mathcal{V}\right\}, \tag{32}$$

where $\mathcal{V} = \{\boldsymbol{u}_1^s, \ldots, \boldsymbol{u}_N^s\}$ is a collection of simulated $\boldsymbol{u}_j^s \sim N(\boldsymbol{0}, \boldsymbol{I})$, $j = 1, \ldots, N = |\mathcal{V}|$. Therefore, as in Theorem 4 and using (30), we define a modified confidence set for τ_0 as

$$\Xi'_{1-\alpha}(\boldsymbol{y}_{obs}) = \big\{\tau : \min_{(\tau, \boldsymbol{M}_\tau) \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})} \widetilde{T}(\boldsymbol{y}_{obs}, (\tau, \boldsymbol{M}_\tau)) \leq 1 - \alpha\big\}. \tag{33}$$

Theorem 6 below states that $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{y}_{obs})$ in (32) add $\Xi'_{1-\alpha}(\boldsymbol{y}_{obs})$ in (33) have the desired coverage rates when $|\mathcal{V}|$ is large enough. One complication preventing us from directly applying Theorem 4 here to the Gaussian mixture model is: Condition (N1) may not hold for two events of arbitrarily small probabilities. They are: (i) when $\boldsymbol{u}^{rel}$ hits certain directions (diag$(\boldsymbol{M}_0\boldsymbol{\sigma}_0)\boldsymbol{u}^{rel}$ and $\{I - \boldsymbol{M}_\tau(\boldsymbol{M}_\tau^\top \boldsymbol{M}_\tau)^{-1}\ \boldsymbol{M}_\tau^\top\}\boldsymbol{M}_0\boldsymbol{\mu}_0$ are on the same direction for some $\boldsymbol{M}_\tau \neq \boldsymbol{M}_0$, $\tau \leq \tau_0$); (ii) when $\|\boldsymbol{u}^{rel}\|$ is extremely large. Fortunately, we can control both events so condition (N1) still holds in probability. Thus, $\widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})$ and $\Xi'_{1-\alpha}(\boldsymbol{y}_{obs})$ maintain the desired coverage rates except for an arbitrarily small $\delta > 0$. A proof is in Appendix C.1.3.

Theorem 6. *Suppose $n - \tau_0 > 4$. For any arbitrarily small $\delta > 0$, there exist $\gamma_\delta > 0, c > 0$ and also an interval of positive width $\Lambda_\delta = [\frac{\gamma_\delta^{1.5}}{\log(n)/n}, \frac{\log\{0.5C_{\min}^2\gamma_\delta+1\}}{2\tau_0\log(n)/n}]$ such that when $\lambda \in \Lambda_\delta$, (a) $\mathbb{P}_{\boldsymbol{U}|\mathcal{V}}\{(\tau_0, \boldsymbol{M}_0) \in \widehat{\Upsilon}_{\mathcal{V}}(\boldsymbol{Y})\} \geq 1 - \delta - o_p(e^{-c|\mathcal{V}|})$, and (b) $\mathbb{P}_{\boldsymbol{U}|\mathcal{V}}\{\tau_0 \in \Xi'_{1-\alpha}(\boldsymbol{Y})\} \geq 1 - \alpha - \delta - o_p(e^{-c|\mathcal{V}|})$. Here, a random sample $\boldsymbol{Y}$ is generated with the set $\boldsymbol{\theta} = \boldsymbol{\theta}_0 = (\tau_0, \boldsymbol{M}_0, \boldsymbol{\mu}_0, \boldsymbol{\sigma}_0)$. Also, $\Xi'_{1-\alpha}(\cdot)$ is defined in (33) and $C_{\min} = \min_{\{\boldsymbol{M}_\tau : |\tau| \leq |\tau_0|, \tau \neq \tau_0\}} \|(\boldsymbol{I} - \boldsymbol{M}_\tau)\boldsymbol{M}_0\boldsymbol{\mu}_0\| > 0$.*

Theorem 6 states that we can recover both τ_0 and $\boldsymbol{M}_0$ with a high certainty, if we have enough computing resource to allow for a large $|\mathcal{V}|$. Nevertheless, getting each element in $\boldsymbol{M}_0$ correctly is a much harder task than recovering τ_0, requiring a much larger $|\mathcal{V}|$. This is because, if an alternative $\boldsymbol{M}_\tau$ is not too different from $\boldsymbol{M}_0$, a repro sample $\boldsymbol{y}^s$ generated using the $\boldsymbol{M}_\tau$ can very well produce the same τ estimate as that generated by using $\boldsymbol{M}_0$. In practice, our computing power is limited so we focus only the task of making inference for τ_0. Our numerical studies in Section 5 suggest that the confidence set of τ_0 in (33) can achieve its desirable coverage rate empirically without $|\mathcal{V}|$ being excessively large.

Finally, to gain further insight, we provides below an upper bound for $\tau_{up}(\boldsymbol{y}_{obs}) = \sup\{\tau : \exists \boldsymbol{M}_\tau,\ \text{s.t.}\ (\tau, \boldsymbol{M}_\tau) \in \widehat{\Upsilon}_{\mathcal{V}}\}$, the largest τ that can be included in the candidate set $\widehat{\Upsilon}_{\mathcal{V}}$.

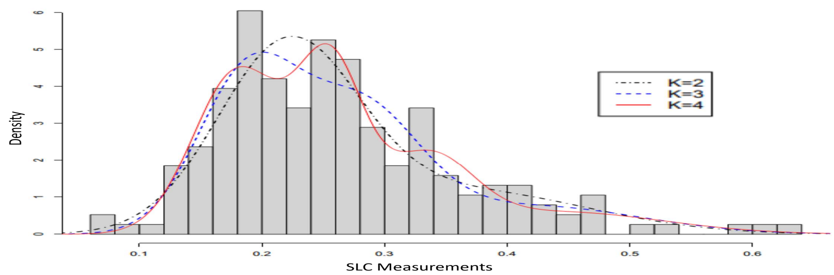

Figure 1: Histogram of 190 SLC measurements and estimated Gaussian mixture density curves.

Theorem 7. *If $\lambda = O(n/\log(n))$ as in Theorem 6, there exist constants a, a' s.t. $\mathbb{P}(\tau_{up}(\boldsymbol{Y}) \geq \tau) \leq a \exp\{-a'(n^{O((\tau-\tau_0)/\log(n))-1/2} - n^{1/2}) + \log|\mathcal{V}|\}$, where $\tau \geq \tau_0$.*

When $\tau - \tau_0 \geq O(\log(n))$, the probability bound decays fast with n, indicating the upper bound of the confidence set is $\tau_0 + O(\log(n))$ with large probability.

4.3 Inference for the component mean $\mu_j^{(0)}$ and standard deviation $\sigma_j^{(0)}$

We use the three-step procedure in Section 3.3 to conduct inference for $\mu_j^{(0)}$ and $\sigma_j^{(0)}$, while accommodating the uncertainty of the unknown τ_0. To mitigate the impact of potential membership mislabeling issues in the mixture, more robust nonparametric quantile estimators are also employed in Steps 2-3. Due to space limit, details are provided in Appendix C.3.

5 Numerical studies: real and simulated data

We conduct a comprehensive numerical study using both real and simulated data. The real data consist of activity measurements of red blood cell sodium–lithium countertransport (SLC) from 190 individuals (Dudley et al., 1991). The SLC measurement is known to be correlated with blood pressure and is considered by many researchers to be an important contributor to hypertension. The same SLC data set has previously been analyzed using Gaussian mixture models with an unknown number of components by Roeder (1994) and Chen et al. (2012).

We apply the repro samples method to the SLC data and obtain more comprehensive and informative inference results. The simulation studies are under two settings that mimic the SLC data and demonstrate the superior empirical performance of our proposed method. Due to space limit, in the main text we present inference results for the discrete parameter τ_0 only. Comparisons with existing approaches, along with additional inference results for $\mu_j^{(0)}$ and $\sigma_j^{(0)}$, are summarized in the main text, with full details provided in Appendix D.

5.1 Inference on the discrete unknown number of components τ_0

Roeder (1994) and Chen et al. (2012) used the classical framework to test the unknown number of components $H_0 : \tau_0 = k$ vs $H_1 : \tau_0 > k$ for a small integer k. Roeder (1994) concluded that $\tau_0 = 3$ is the smallest value of τ_0 not rejected by the data under the assumption that σ_j^2's are the same. Chen et al. (2012) concluded instead the smallest $\tau_0 = 2$ without the equal variance assumption. Our first goal is to obtain a 95% confidence set for the discrete unknown τ_0. To do so we first obtain a candidate set for $(\tau_0, \boldsymbol{M}_0)$. Particularly, for each τ, we solve the objective function in (31) via fitting a mixture linear regression model with $\boldsymbol{y}_{obs}$ as the response vector and $\boldsymbol{u}^*$ as the covariate using the R package *flexmix* (Grün and Leisch, 2007), and we get $(\tau^*, \widehat{M}_\tau^*)$ by (31). The collection of these $(\tau^*, \widehat{M}_\tau^*)$'s forms a candidate set $\widehat{\Upsilon}(\boldsymbol{y}_{obs})$. To evaluate the nuclear mapping in (28), we set $K_{\max} = \tau_{up}(\boldsymbol{y}_{obs}) \vee 10$, with $\tau_{up}(\boldsymbol{y}_{obs})$ defined in Theorem 7. We then proceed and use formula (33) to obtain a 95% confidence set for τ_0, which is $\Xi'_{1-\alpha}(\boldsymbol{y}_{obs}) = \{2, 3, 4\}$. The confidence set suggests that both $\tau_0 = 2$ and 3 are plausible, consistent with earlier studies. In genetics, a two-component mixture corresponds to a simple dominance model and a three-component corresponds to an additive model (Roeder, 1994). Evidently, we cannot rule out these two models based on the SLC data alone. Our confidence set also includes $\tau_0 = 4$. Figure 1 is the histogram of the SLC measurements together with estimated Gaussian mixture densities with $\tau = 2, 3, 4$ using an EM algorithm (R package *ClusterR*). Notably, only the model with $\tau = 4$ captures all the three spikes in the histogram, while smoothly fitting the rest of the histogram. Therefore, the model with $\tau_0 = 4$ also demands further investigation, as it appears to well represent the data empirically. An advantage of our method is that our confidence set provides both lower and upper bounds, for plausible τ_0 values supported by the data, whereas inverting the test methods proposed by Roeder (1994) and Chen et al. (2012) can only give us one-sided confidence sets.

We use simulation studies to further demonstrate the performance of our proposed method, where the simulation settings mimic that of the real SLC data. Specifically, the true parameters of our simulation models are those in Table A4(a) (from fitting SLC data) with $\tau = 3$ and 4, respectively. In each setting of $\tau_0 = 3$ or 4, we simulate $n = 190$ data points using the corresponding parameters. For each simulated data, we apply the proposed repro samples method. The experiments are repeated 200 times in each setting of $\tau_0 = 3$ or 4.

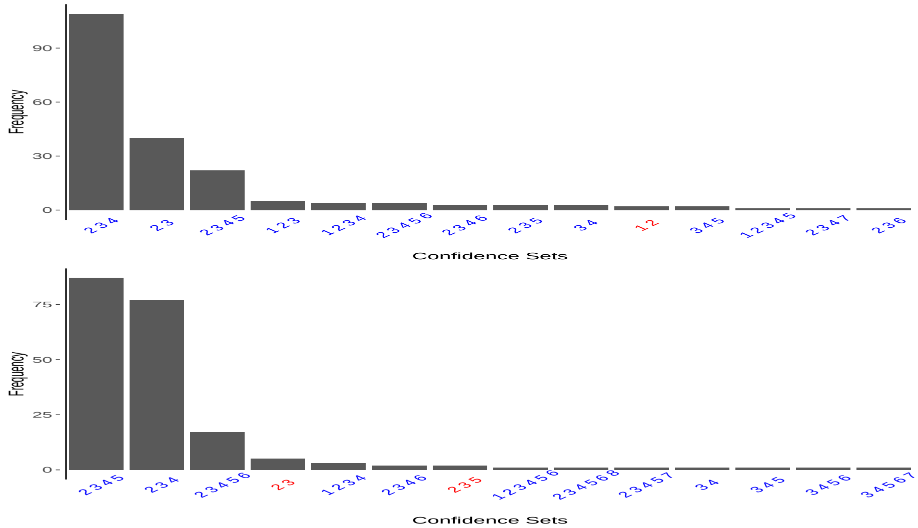

Figure 2: Bar plots of 200 level-95% repro samples confidence sets obtained in the simulation study (200 repetitions). The upper panel is for true $\tau_0 = 3$; lower for true $\tau_0 = 4$.

The bar plots in Figure 2 summarize the 200 level-95% confidence sets of τ_0 obtained in the 200 repetitions using the repro samples method for $\tau_0 = 3$ and 4, respectively. The results demonstrate good empirical performances of the proposed confidence set in both settings, with the τ_0 coverage rate 99% and 96.5%, respectively. Over-coverage is observed but also expected, since the parameter space of τ_0 is discrete and it is often impossible to achieve the exact level of 95%. In the case of $\tau_0 = 3$, the average size of the proposed confidence set is about 3 (standard error 0.05), with the set $\{2, 3, 4\}$ being the majority among the 200 simulations. As for $\tau_0 = 4$, more than 80% of the times, a confidence set is either $\{2, 3, 4\}$ or $\{2, 3, 4, 5\}$, with the average size of the confidence set around 3.65 (standard error 0.05).

5.2 Comparisons to existing approaches and additional inference on $\mu_j^{(0)}$ and $\sigma_j^{(0)}$

We have conducted a comprehensive study to compare the numerical performance of our method with that of several existing frequentist and Bayesian approaches using both the SLC real data and the same 2×200 simulated data sets. Due to space limit, we only state our conclusion here and the detailed results are reported in Appendix D.1. First, point estimators perform poorly in our simulation studies that mimic the SLC data (with overlapping components). For instance, the BIC point estimator, although consistent theoretically, rarely recovers the true τ_0: It correctly identifies the true value of τ_0 only 6.5% of the time for the setting of $\tau_0 = 3$ and never for the setting of $\tau_0 = 4$. This underscores the critical importance of quantifying the estimation uncertainty through a confidence set for τ_0. Second, the approaches developed under the classical test framework using (modified) penalized likelihood ratio tests (e.g., Roeder, 1994; Chen et al., 2012) perform only a one-sided test; thus it is often proposed to adopt "the smallest τ_0 that can not be rejected." We find in our studies the method by Chen et al. (2012) often inaccurately favors smaller models, incorrectly maintaining $H_0 : \tau_0 = 2$ 74% of the time when actually $\tau_0 = 3$, and failing to reject $H_0 : \tau_0 = 3$ more than 93% of the time when actually $\tau_0 = 4$. The results highlight the tendency of these one-sided tests to erroneously prefer a wrong τ over the correct one. Third, the performance of Bayesian methods (e.g., Richardson and Green, 1997) is highly sensitive to the choices of priors on τ, $\boldsymbol{\mu}$, and $\boldsymbol{\sigma}$, and the credible sets do not have frequentist coverage rates. See Appendix D.1 for more details.

We also apply the three-step procedure in Section 3.3 to conduct inference for the location and scale parameters $\mu_j^{(0)}$ and $\sigma_j^{(0)}$. Since Step 1 has already been completed in Section 5.1, we proceed directly to Steps 2–3. In the real data analysis, the proposed method yields stable and interpretable confidence intervals for both parameters, providing a coherent joint quantification of uncertainty across discrete and continuous components. The simulation studies further show that our method is the only approach that consistently achieves the desired empirical coverage for both means and variances, highlighting its robustness and practical reliability, particularly in finite samples. Detailed results are reported in Appendix D.2.

Based on the numerical results and comparisons in Sections 5.1–5.2, we can see that the repro samples method is the only method that gives us a two-sided confidence set for τ_0 with the desired performance. The repro samples method can also effectively quantify uncertainty and makes joint inferences for both the discrete τ_0 and the continuous $\mu_j^{(0)}$ and $\sigma_j^{(0)}$, $j = 1, \ldots, \tau_0$. In summary, the repro samples method gives us an effective approach for analyzing Gaussian mixture data. It provides a performance-guaranteed comprehensive solution for a set of complex irregular inference problems, while the existing procedures can not.

6 Concluding remark and further discussions

We have developed an effective and flexible new framework for inference by utilizing, explicitly or implicitly, synthetic data that mimic the observed sample. A notable strength is that it directly uses inversion techniques and does not rely on large-sample theory or likelihood functions, making it particularly effective for some previously unaddressed challenging problems, including those involving discrete or non-numerical parameters or non-numerical data. By accommodating diverse parameters, models, and data types, it can greatly extend the scope of statistical inference. We also compare the proposed method with the classical Neyman–Pearson framework. When the same test statistic is used as a nuclear mapping, the repro samples method provides the same or better inference results than those obtained via Neyman inversion. The case study of Gaussian mixture illustrates the method's effectiveness in resolving the long-standing challenging inference problem of the unknown number of components, and numerical studies further show that the proposed method is the only approach that can achieve the desired coverage rates for both discrete and continuous parameters.

The generative model specification (1) may not be unique — the same probabilistic model (i.e., the same likelihood) can possibly be represented through different generative models (Hannig, 2009, Remark 6). For example, the uniform model $Y_i \sim U(\theta, \theta^2)$, $\theta > 1$, may be expressed in two different forms of (1), one using the full data and the other using the minimal sufficient statistics (the sample minimum and maximum) (Hannig et al., 2016, Example 4). Within the repro samples framework, these different model formulations still lead to identical inference on θ if the same nuclear mapping is used; see Example A6 in Appendix E.3.2. This contrasts with other Fisher inversion-based GFI and IM approaches, in which the resulting inference depends on the distributional form of the latent U's and thus on the chosen generative model; cf., Hannig et al. (2016); Martin and Liu (2013).

Our primary developments are under the generative model (1), but can be extended to more general settings, such as (11) that only requires the target (true or oracle) parameter be recovered by a given algorithm. Nevertheless, all models are approximations, making it important to consider misspecification. We have begun extending the repro samples method to settings with model misspecification, where a family of working models is used to analyze data generated from a completely unknown model. For example, in Hou et al. (2025) on high-dimensional data classification, although the true model is completely unknown, inference is conducted using misspecified sparse working models, targeting oracle parameters that minimize an appropriate loss function and represent the best working-model (closest to observed data). The development is a further extension of (11) and aligns with several relevant research in model calibration, variational Bayes, robust universal inference, robust SBI, among others (e.g., Smith, 2007; Wang and Blei, 2019; Park et al., 2023; Tomaselli et al., 2025).

Finally, despite the two techniques in Section 3.4, the computational cost can remain high for complex inference problems. For example, obtaining candidate sets for discrete parameters may require repeated runs of the inversion algorithm, with a cost comparable to bootstrap. This poses further challenges when an effective inversion algorithm is unavailable. Nevertheless, with continued growth in computing power, we view this as a characteristic shared by many modern simulation-based approaches rather than a fundamental limitation.

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